

VIETNAM NATIONAL UNIVERSITY-HO CHI MINH CITY

HO CHI MINH UNIVERSITY OF SCIENCE

FACULTY OF INFORMATION TECHNOLOGY

KNOWLEDGE ENGINEERING DEPARTMENT

Self-Study Course Report

Program: Machine Learning Specialization

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1 Introduction

1.1 About the Program

- **Program Platform:** Coursera
- **Program Name:** Machine Learning Specialization
- **Organization:** DeepLearning.AI and Stanford Online
- **Program Link:** [Machine Learning Specialization](#)
- **Courses in Program:**
 1. *Supervised Machine Learning: Regression and Classification*
 2. *Advanced Learning Algorithms*
 3. *Unsupervised Learning, Recommenders, and Reinforcement Learning*
- **Duration:** From 17 October, 2025 to 24 November, 2025

1.2 Details of Registration

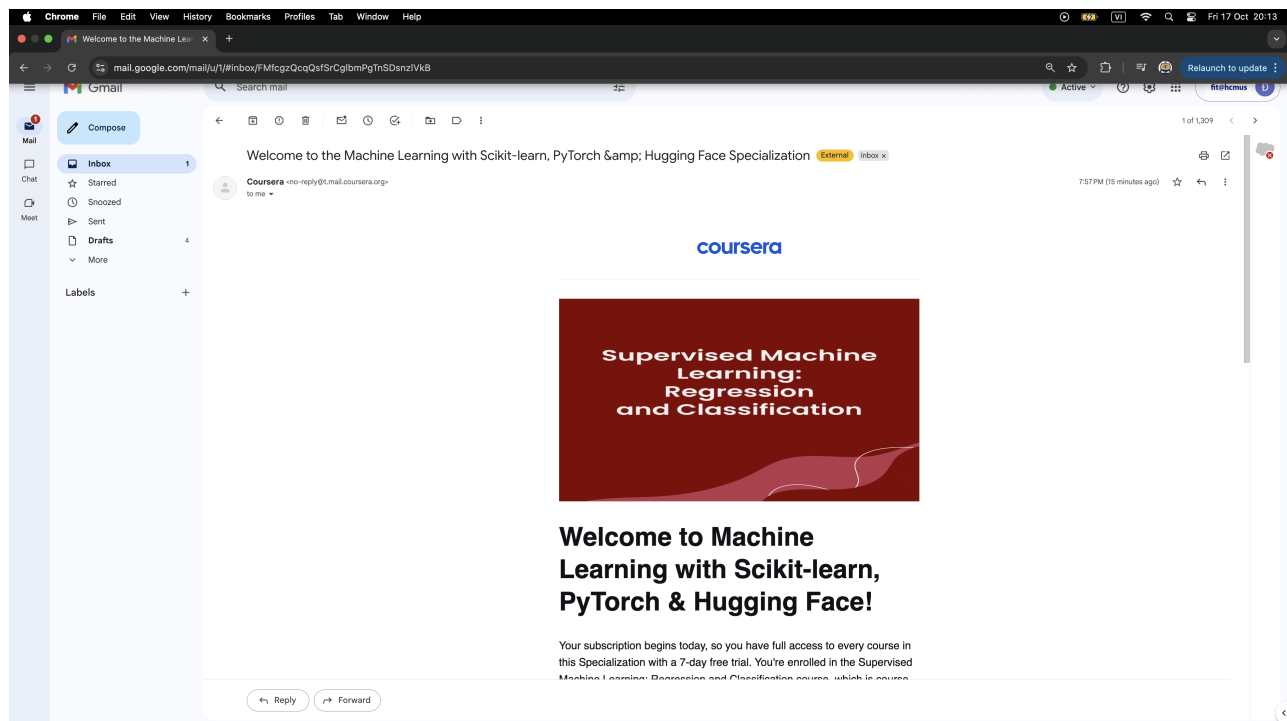


Figure 1: Registration confirmation on Coursera (17/10/2025)

2 Overview

2.1 Details of the Program

The **Machine Learning Specialization** is a foundational online program created in collaboration between **DeepLearning.AI** and **Stanford Online**. This beginner-friendly specialization teaches the fundamentals of machine learning and how to apply these techniques to build real-world AI applications.

This specialization is taught by **Andrew Ng**, an AI visionary who has led pioneering research at Stanford University and groundbreaking work at Google Brain, Baidu, and Landing.AI to advance the field of artificial intelligence.

This 3-course specialization is an updated version of Andrew Ng's pioneering *Machine Learning* course, rated 4.9 out of 5 and taken by over 4.8 million learners since it launched in 2012.

It provides a broad introduction to modern machine learning, including:

- **Supervised Learning:** multiple linear regression, logistic regression, neural networks, and decision trees.
- **Unsupervised Learning:** clustering, dimensionality reduction, and recommender systems.
- **Best Practices:** evaluating and tuning models, and taking a data-centric approach to improving performance.

By the end of this specialization, learners will have mastered key concepts and gained the practical know-how to powerfully apply machine learning to challenging real-world problems. If you're looking to break into AI or build a career in machine learning, this specialization is the ideal place to start.

2.2 Projects in Specialization - 3 course series

- Build machine learning models in Python using popular machine learning libraries NumPy and scikit-learn.
- Build and train supervised machine learning models for prediction and binary classification tasks, including linear regression and logistic regression.
- Build and train a neural network with TensorFlow to perform multi-class classification.
- Apply best practices for machine learning development so that your models generalize to data and tasks in the real world.
- Build and use decision trees and tree ensemble methods, including random forests and boosted trees.
- Use unsupervised learning techniques for unsupervised learning: including clustering and anomaly detection.
- Build recommender systems with a collaborative filtering approach and a content-based deep learning method.
- Build a deep reinforcement learning model.

3 Supervised Machine Learning: Regression and Classification

This foundational course established the core principles of machine learning. It provided a comprehensive journey from the mathematical formulation of learning problems to the implementation of gradient-based optimization for both continuous (regression) and discrete (classification) prediction tasks.

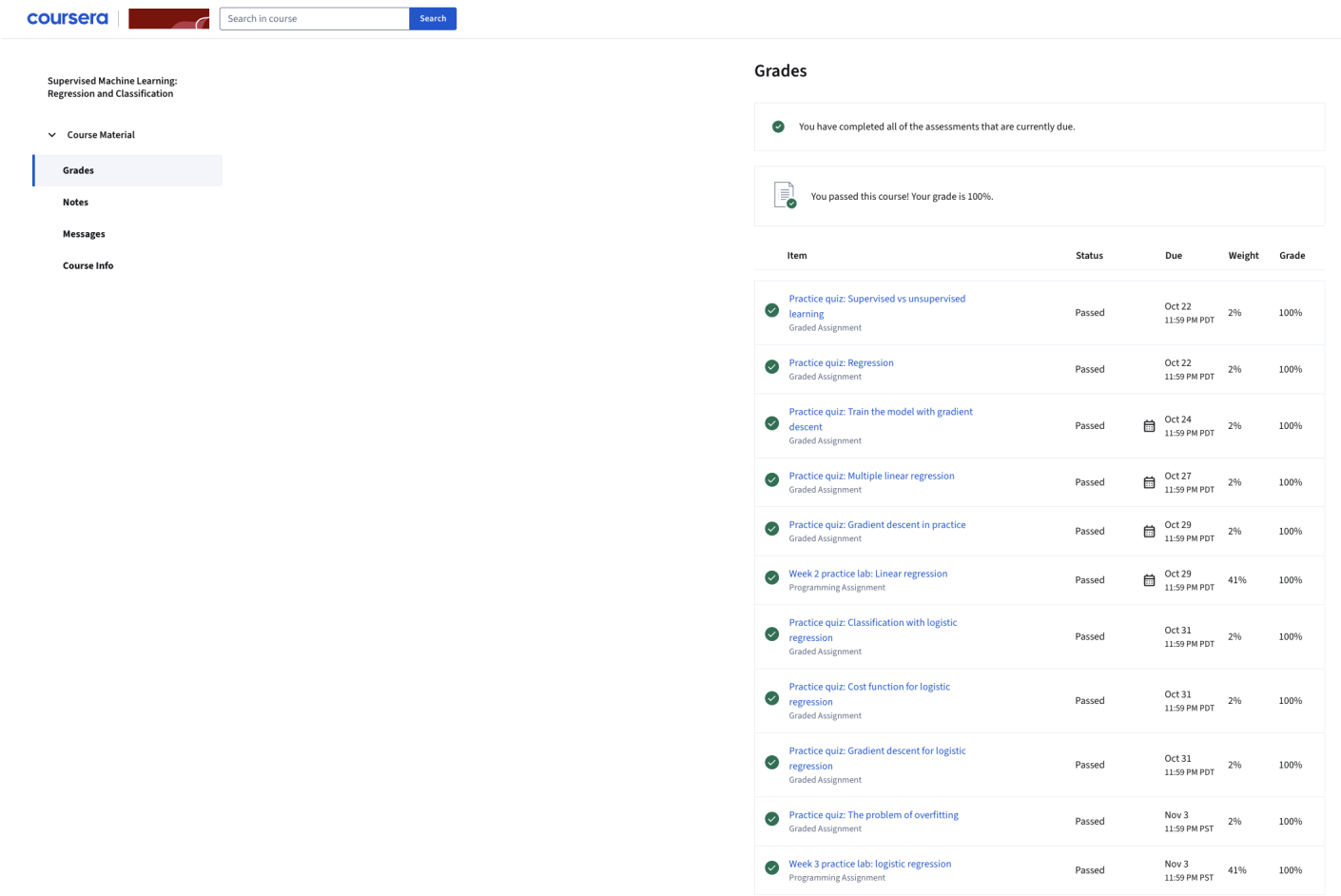


Figure 2: Course 1 completed

3.1 Week 1: Introduction to Machine Learning

3.1.1 Key Insights and Learnings

This module introduced the fundamental paradigm shift from traditional programming to data-driven learning.

- **The Supervised Learning Framework:** Structured the problem-solving approach around three core components:
 1. **Model Representation:** Mathematically defined the relationship between inputs x and output y (e.g., Linear Regression $f_{w,b}(x) = wx + b$).

2. **Cost Function (J):** Quantified the model's error. For regression, the **Mean Squared Error (MSE)** is used to measure the average squared difference between predictions and actual targets.
3. **Gradient Descent Optimization:** Implemented an iterative algorithm that updates parameters (w, b) by moving against the gradient of the cost function ($\alpha \frac{\partial J}{\partial w}$) to reach the global minimum.

- **Learning Paradigms:**

- **Supervised Learning:** Training models on labeled data (Input $X \rightarrow$ Output Y) to predict future outcomes.
- **Unsupervised Learning:** Extracting hidden structures or clusters from unlabeled data (explored deeply in later courses).

3.1.2 Learning Evidence

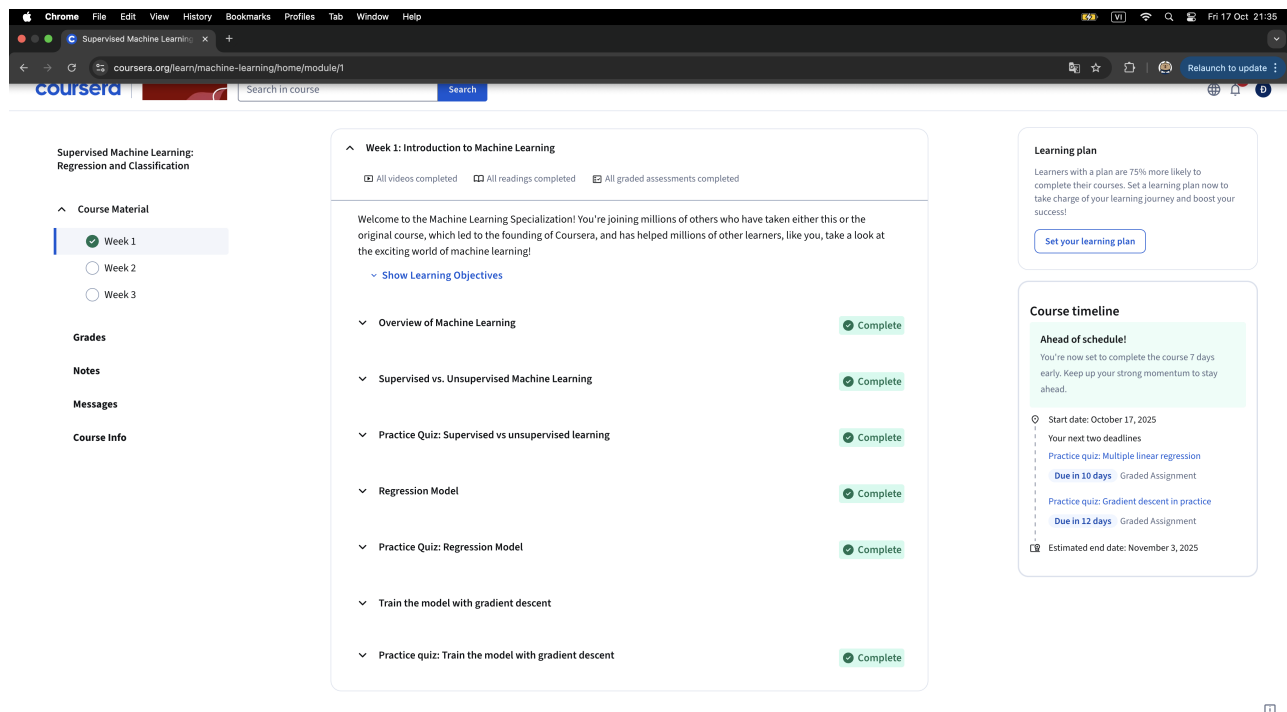


Figure 3: Lesson's completed date (17/10/2025)

3.1.3 Graded Assignment Evidence

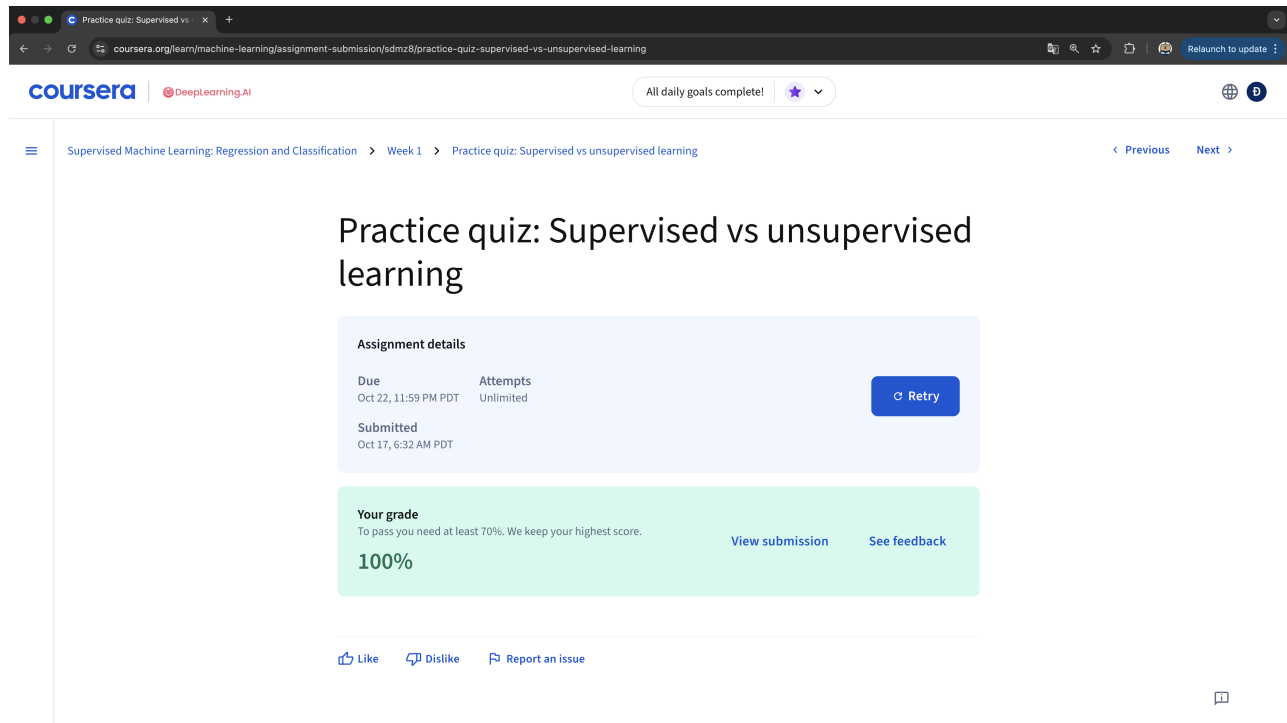


Figure 4: Completed Graded Assignment 1 (Supervised vs unsupervised learning)

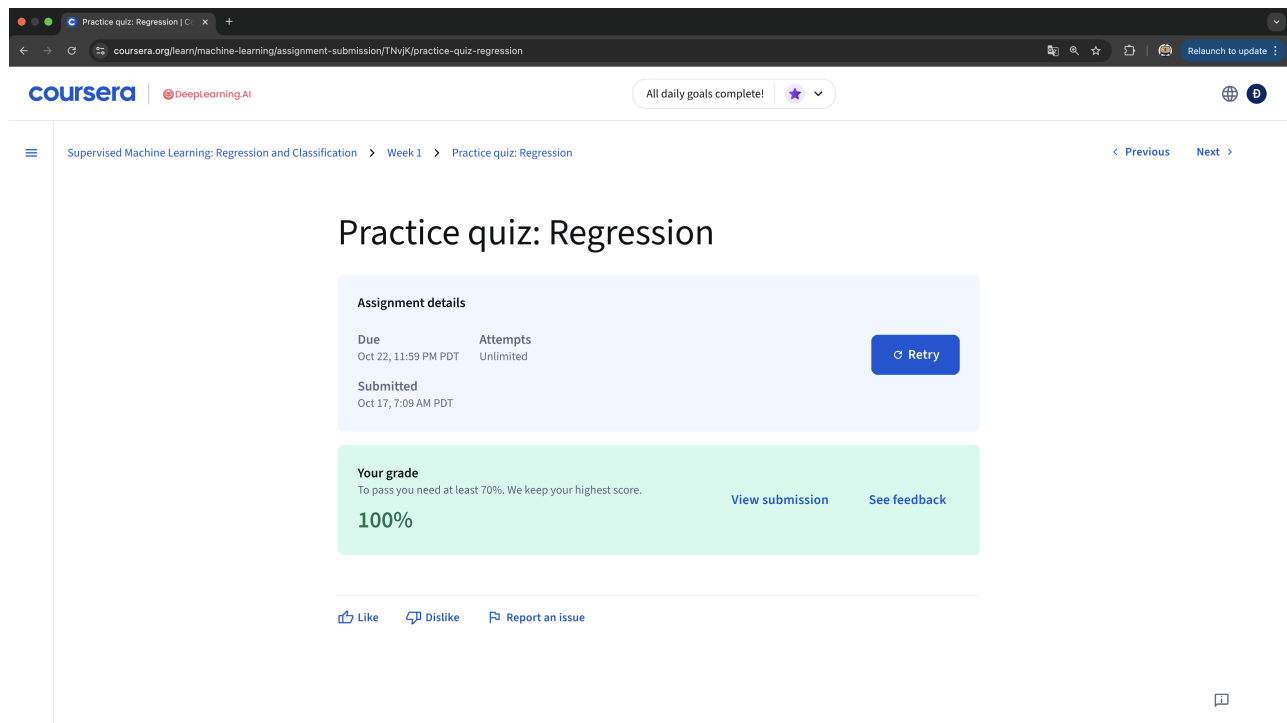


Figure 5: Completed Graded Assignment 2 (Regression)

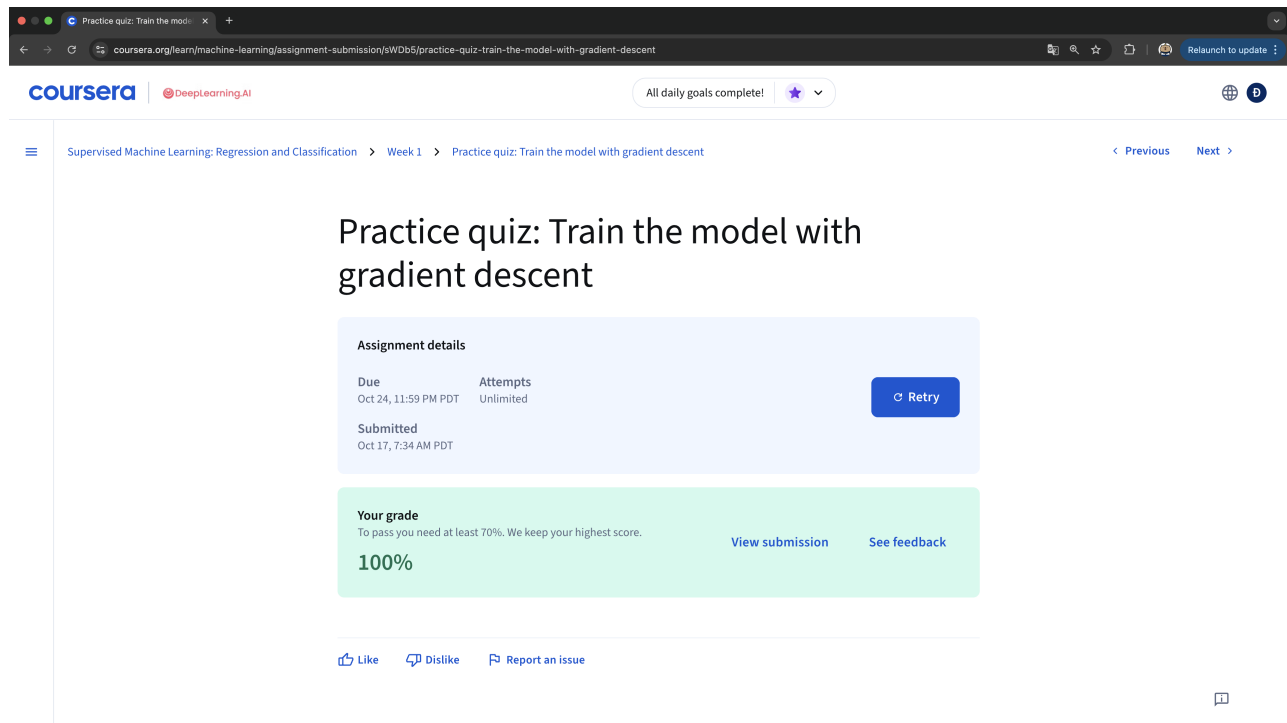


Figure 6: Completed Graded Assignment 3 (Train the model with gradient descent)

3.2 Week 2: Regression with Multiple Input Variables

3.2.1 Key Insights and Learnings

This module expanded linear regression to handle complex, real-world datasets with multiple features, focusing on computational efficiency and convergence speed.

- **Vectorization (NumPy):** Leveraged linear algebra principles to process large datasets. By replacing explicit `for` loops with vector/matrix operations (`np.dot`), computation time is drastically reduced, enabling the training of scalable models.
- **Feature Scaling & Convergence:**
 - **Problem:** When features have vastly different scales (e.g., price vs. number of bedrooms), the cost function contours become elongated (elliptical), causing Gradient Descent to oscillate and converge slowly.
 - **Solution:** Applied **Z-score Normalization** to rescale features. This transforms the cost function contours into circles, allowing Gradient Descent to take a direct path to the minimum.
- **Feature Engineering and Polynomial Regression:** Learned that model power isn't limited by linear features. By engineering new features (e.g., $x^2, x^3, \sqrt{x}$), a linear model can be adapted to fit non-linear curves, enhancing flexibility without changing the underlying optimization algorithm.

3.2.2 Learning Evidence

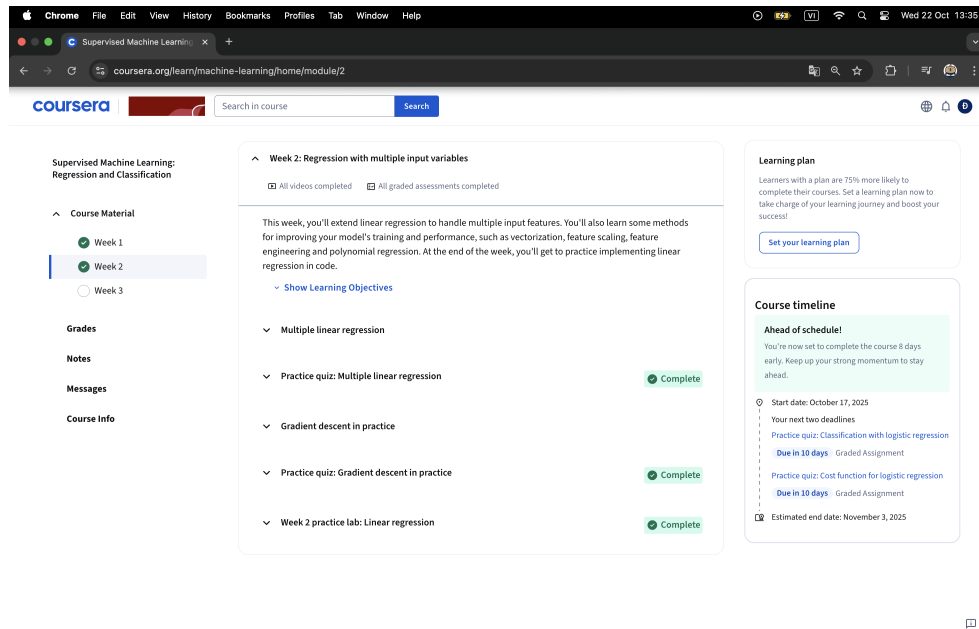


Figure 7: Lesson's completed date (22/10/2025)

3.2.3 Graded Assignment Evidence

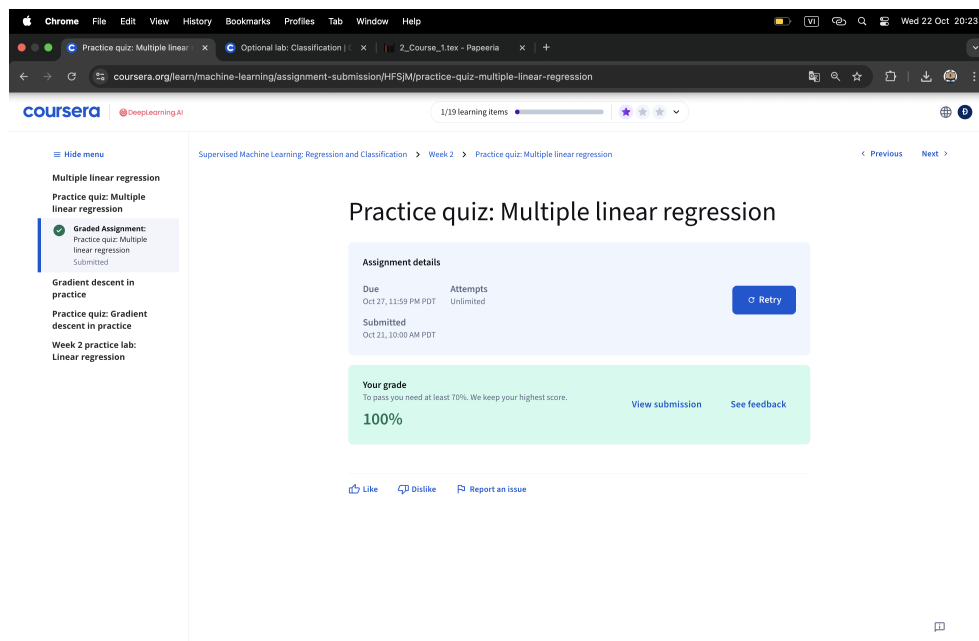


Figure 8: Completed Graded Assignment 1 (Multiple linear regression)

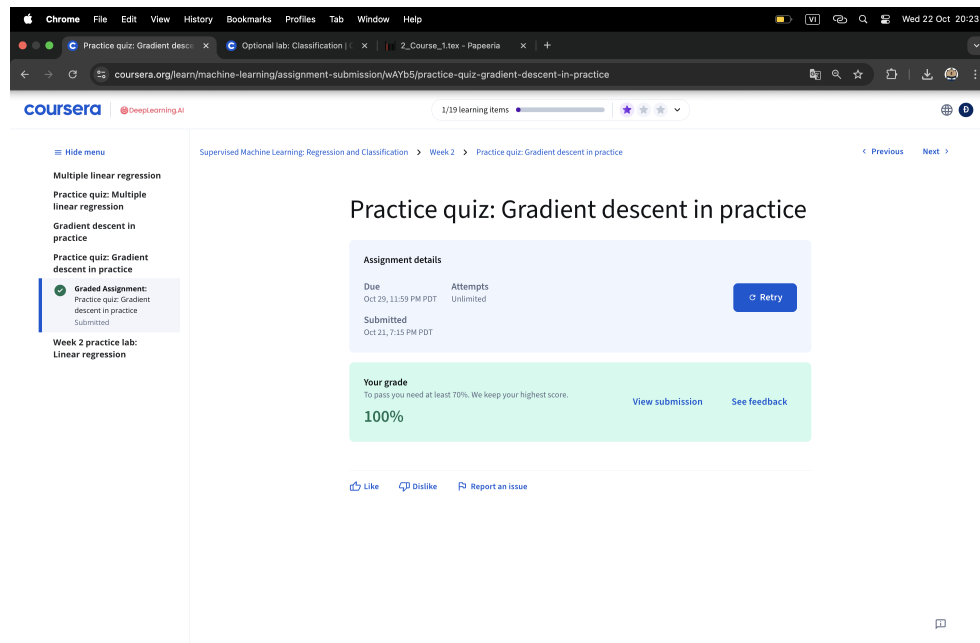


Figure 9: Completed Graded Assignment 2 (Gradient descent in practice)

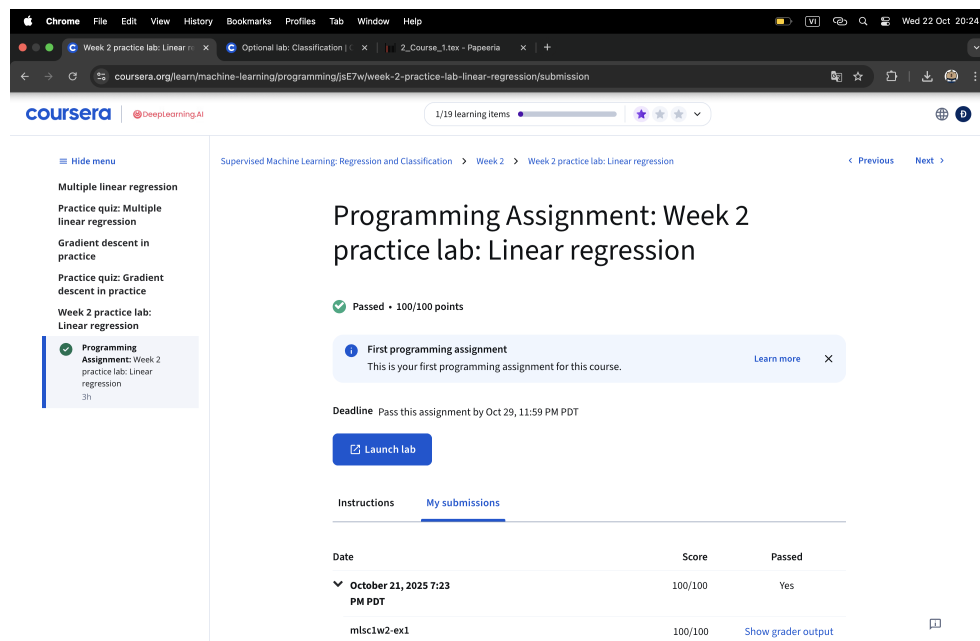


Figure 10: Completed Programming Assignment (Linear regression)

3.3 Week 3: Classification

3.3.1 Key Insights and Learnings

This module introduced the mathematical adaptations required to predict categories rather than continuous numbers.

- **Logistic Regression and The Sigmoid Function:**

- Addressed the limitations of linear regression for classification (unbounded output).
- Applied the **Sigmoid Function** ($g(z) = \frac{1}{1+e^{-z}}$) to map predictions to a probability range $[0, 1]$, interpreting the output as $P(y = 1|x)$.
- **Decision Boundaries:** Visualized classification not just as a prediction task, but as finding a geometric boundary (line or polynomial curve) that separates different classes in the feature space.
- **The Logistic Loss Function:**
 - Learned that using MSE for Logistic Regression results in a **non-convex** cost function (many local minima).
 - Implemented the **Log Loss (Binary Cross-Entropy)** function to ensure the cost function is convex, guaranteeing that Gradient Descent can find the global optimum.
- **Overfitting and Regularization:**
 - **The Problem:** Models that fit training data too closely (high variance) fail to generalize to new data.
 - **The Solution:** Implemented **L2 Regularization**. By adding a penalty term ($\lambda \sum w_j^2$) to the cost function, the model is forced to keep weights small, resulting in a simpler, smoother decision boundary that generalizes better.

3.3.2 Learning Evidence

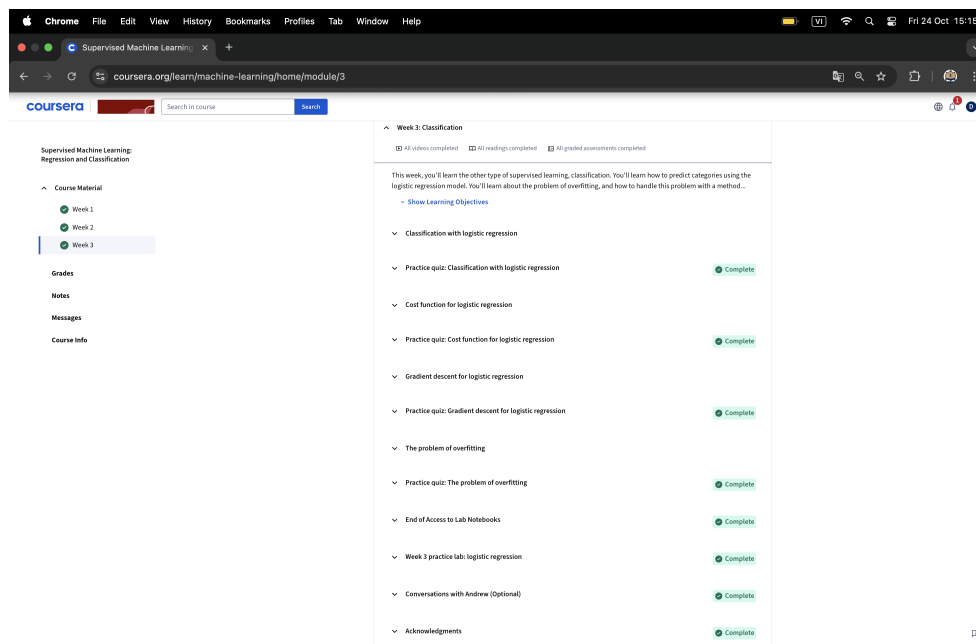


Figure 11: Lesson's completed date (24/10/2025)

3.3.3 Graded Assignment Evidence

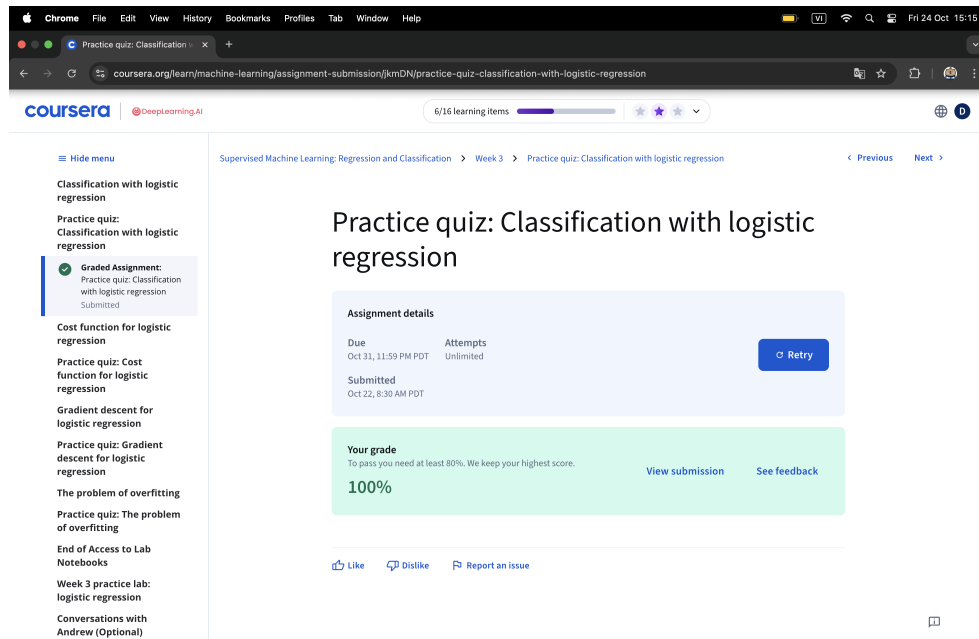


Figure 12: Completed Graded Assignment 1 (Classification with logistic regression)

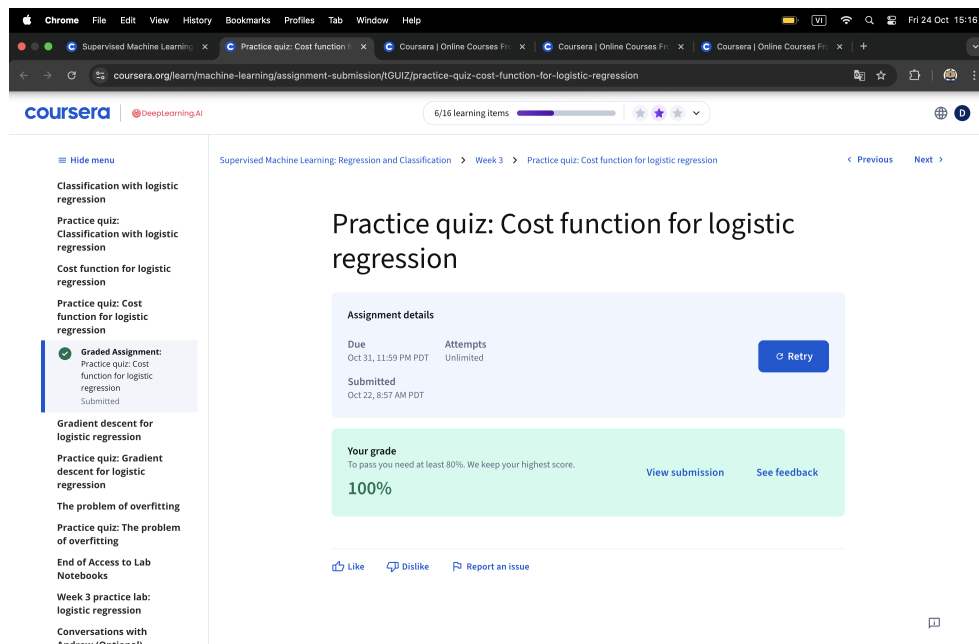


Figure 13: Completed Graded Assignment 2 (Cost function for logistic regression)

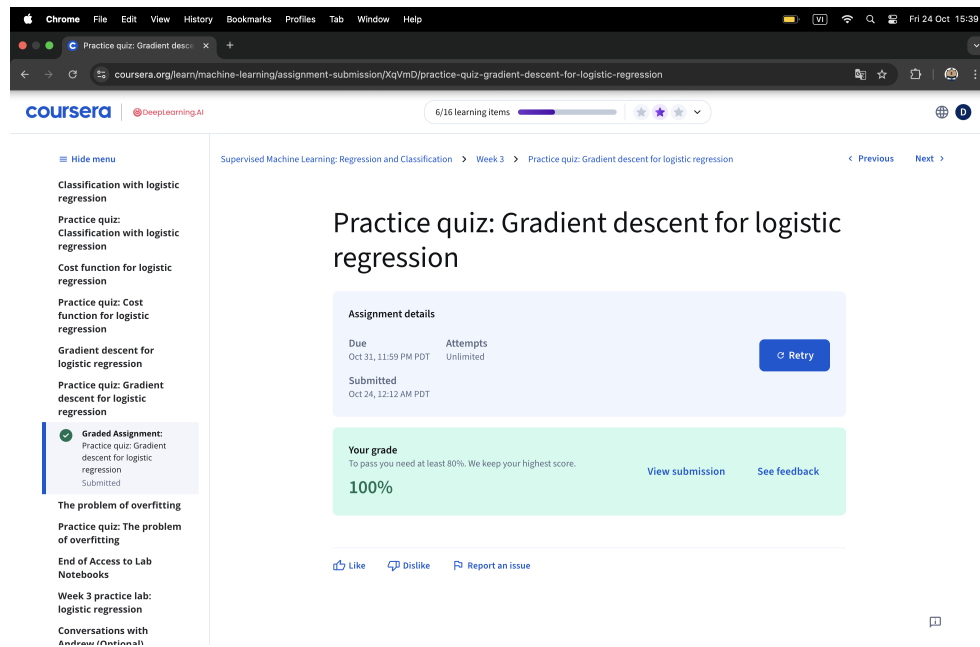


Figure 14: Completed Graded Assignment 3 (Gradient descent for logistic regression)

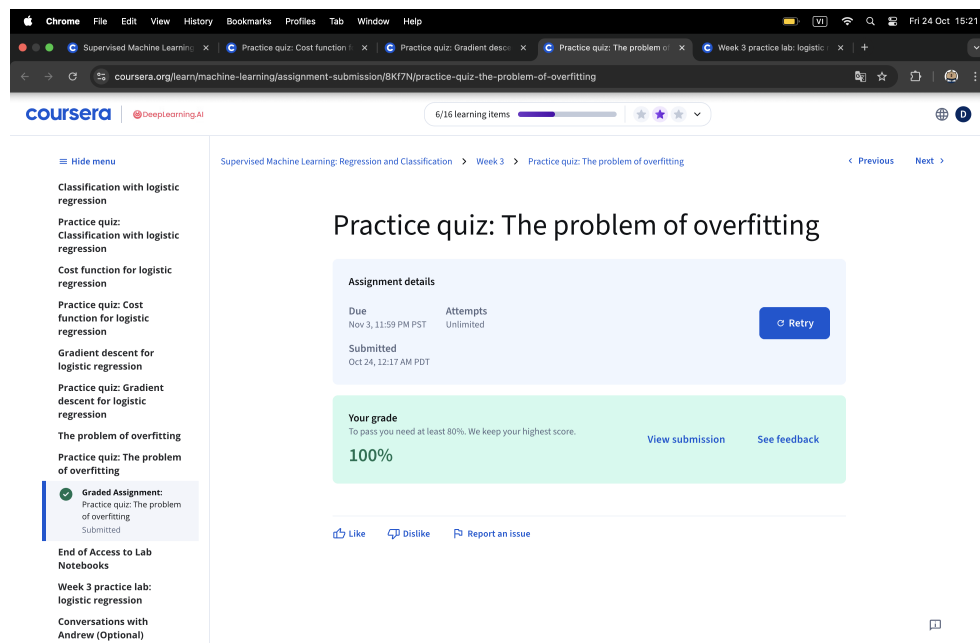


Figure 15: Completed Graded Assignment 4 (The problem of overfitting)

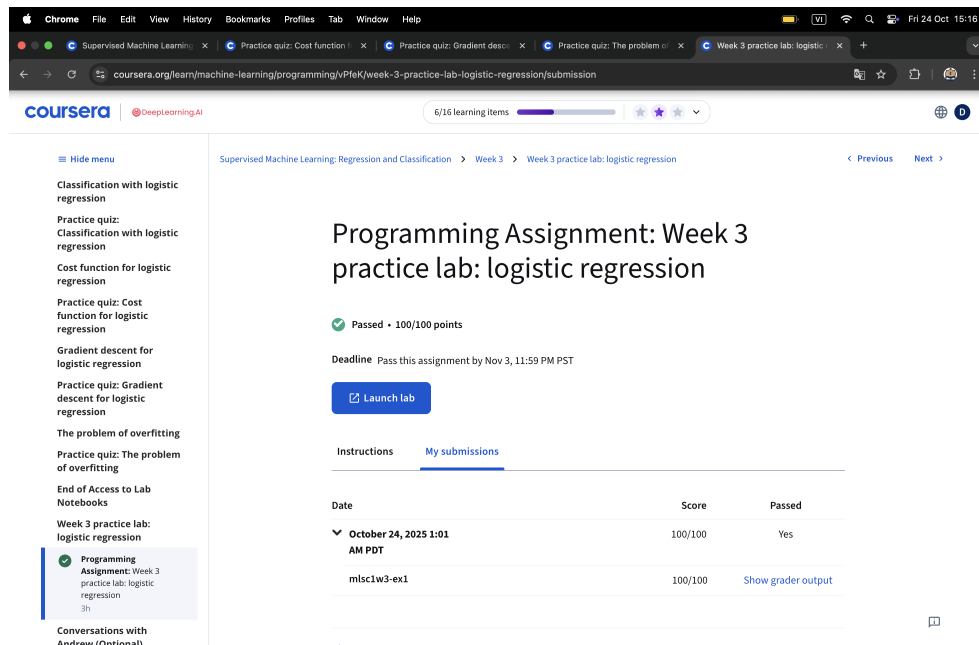
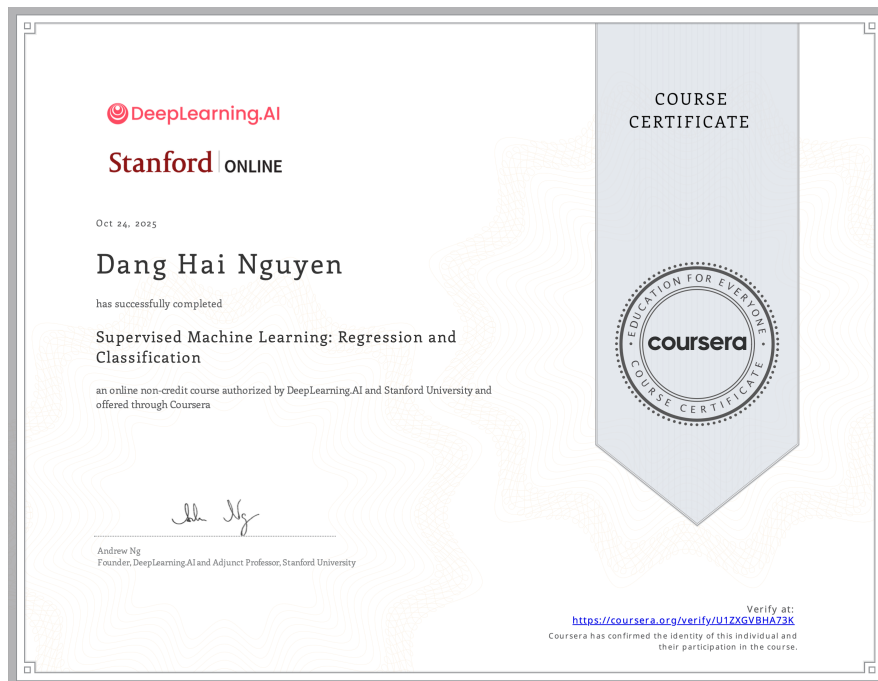


Figure 16: Completed Programming Assignment (Logistic regression)

3.4 Certificate for Completion

Figure 17: Certificate of completion for the course *Supervised Machine Learning: Regression and Classification*.

4 Advanced Learning Algorithms

Building upon the foundational regression and classification models, this course delves into high-capacity models capable of capturing complex, non-linear patterns. It moves from the theoretical underpinnings of Neural Networks to the practical methodology of building, diagnosing, and optimizing real-world machine learning systems.

The screenshot shows the Coursera interface for the 'Advanced Learning Algorithms' course. The 'Grades' section indicates that the user has completed all assessments and passed the course with a 100% grade. A table lists the following items:

Item	Status	Due	Weight	Grade
Practice quiz: Neural networks intuition	Passed	Oct 29 11:59 PM PST	2%	100%
Practice quiz: Neural network model	Passed	Oct 29 11:59 PM PST	2%	100%
Practice quiz: TensorFlow implementation	Passed	Oct 31 11:59 PM PST	2%	100%
Practice quiz: Neural network implementation in Python	Passed	Oct 31 11:59 PM PST	2%	100%
Practice Lab: Neural Networks for Binary Classification	Passed	Nov 3 11:59 PM PST	17%	100%
Practice quiz: Neural Network Training	Passed	Nov 3 11:59 PM PST	2%	100%
Practice quiz: Activation Functions	Passed	Nov 5 11:59 PM PST	2%	100%
Practice quiz: Multiclass Classification	Passed	Nov 5 11:59 PM PST	2%	100%
Practice Lab: Additional Neural Network Concepts	Passed	Nov 7 11:59 PM PST	2%	100%
Practice Lab: Neural Networks for Multiclass classification	Passed	Nov 7 11:59 PM PST	17%	100%
Practice quiz: Advice for applying machine learning	Passed	Nov 10 11:59 PM PST	2%	100%
Practice quiz: Bias and variance	Passed	Nov 12 11:59 PM PST	2%	100%
Practice quiz: Machine learning development process	Passed	Nov 12 11:59 PM PST	2%	100%
Practice Lab: Advice for Applying Machine Learning	Passed	Nov 14 11:59 PM PST	19%	100%

Figure 18: Course 2 completed

4.1 Week 1: Neural Networks

4.1.1 Key Insights and Learnings

This module marked the transition from traditional algorithms to Deep Learning, utilizing biological inspiration to solve complex computational problems.

- **Neural Network Architecture:** Understood how stacking layers of logistic regression units creates a "Multi-Layer Perceptron."
 - **Activation Functions:** Analyzed the role of non-linear functions like **ReLU** (Rectified Linear Unit) for hidden layers to avoid the "vanishing gradient" problem and **Sigmoid** for binary output.
 - **Vectorization:** Leveraged matrix multiplication (via TensorFlow/NumPy) to perform forward propagation efficiently over large datasets.

- **Multiclass Classification (Softmax):** Extended binary classification to multiple categories using the **Softmax Regression** function, which outputs a probability distribution across N classes (ensuring $\sum P(y = k) = 1$).
- **Hierarchical Feature Learning:** Recognized that deep networks learn representations hierarchically—lower layers detect simple edges, while deeper layers assemble them into complex shapes and objects.

4.1.2 Learning Evidence

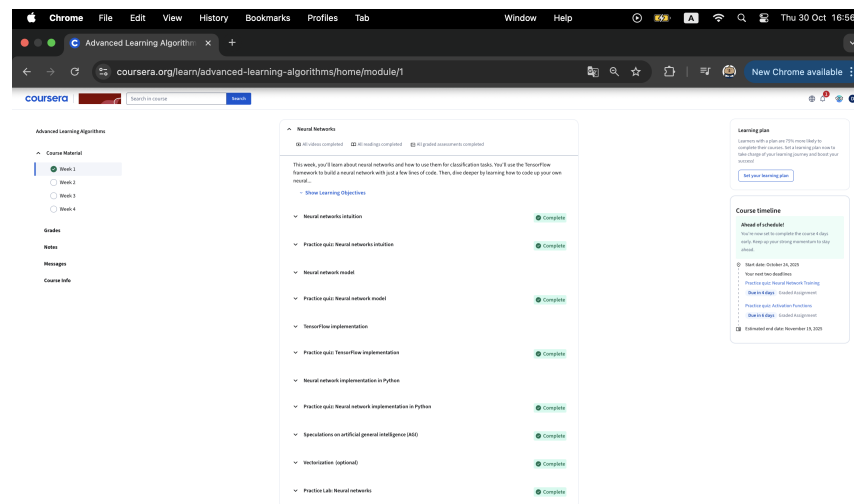


Figure 19: Lesson's completed date (30/10/2025)

4.1.3 Graded Assignment Evidence

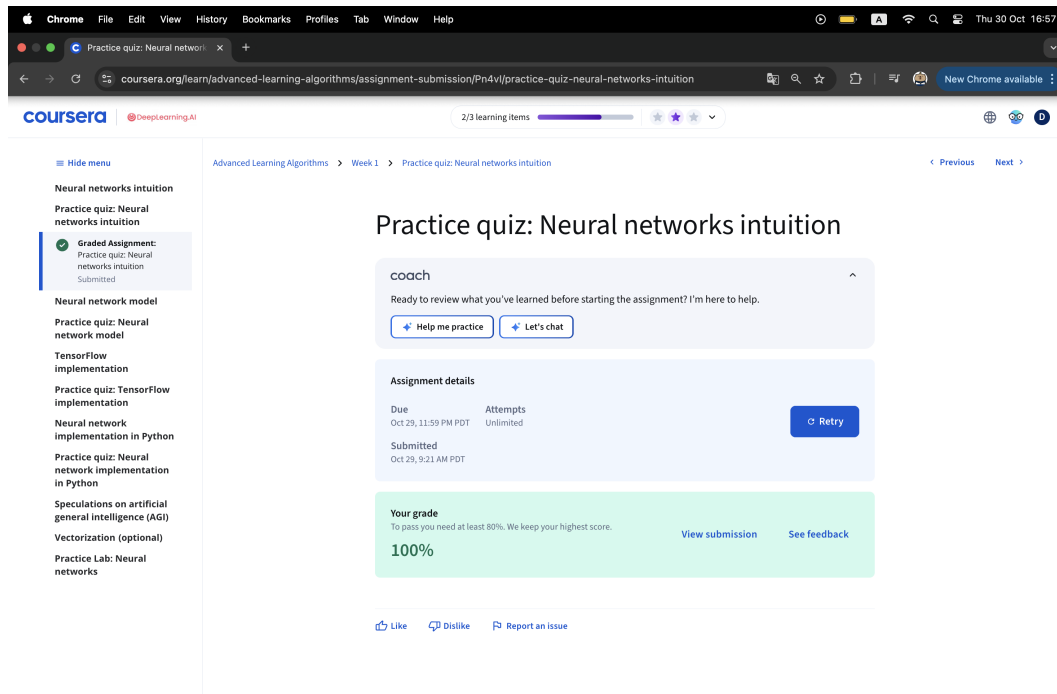


Figure 20: Completed Graded Assignment 1 (Neural networks intuition)

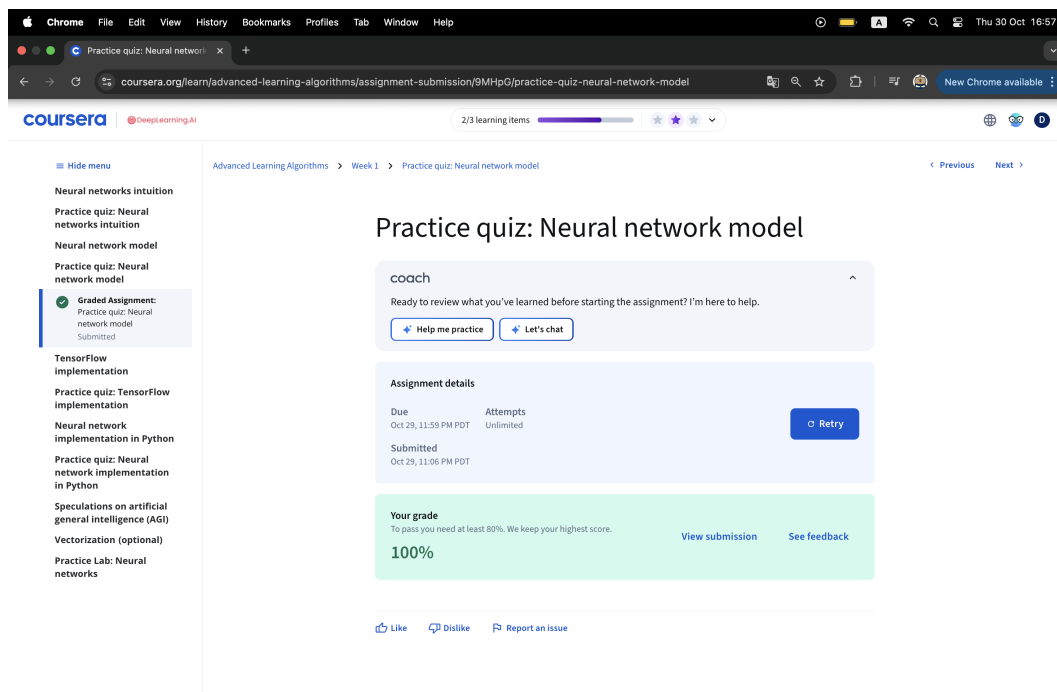


Figure 21: Completed Graded Assignment 2 (Neural network model)

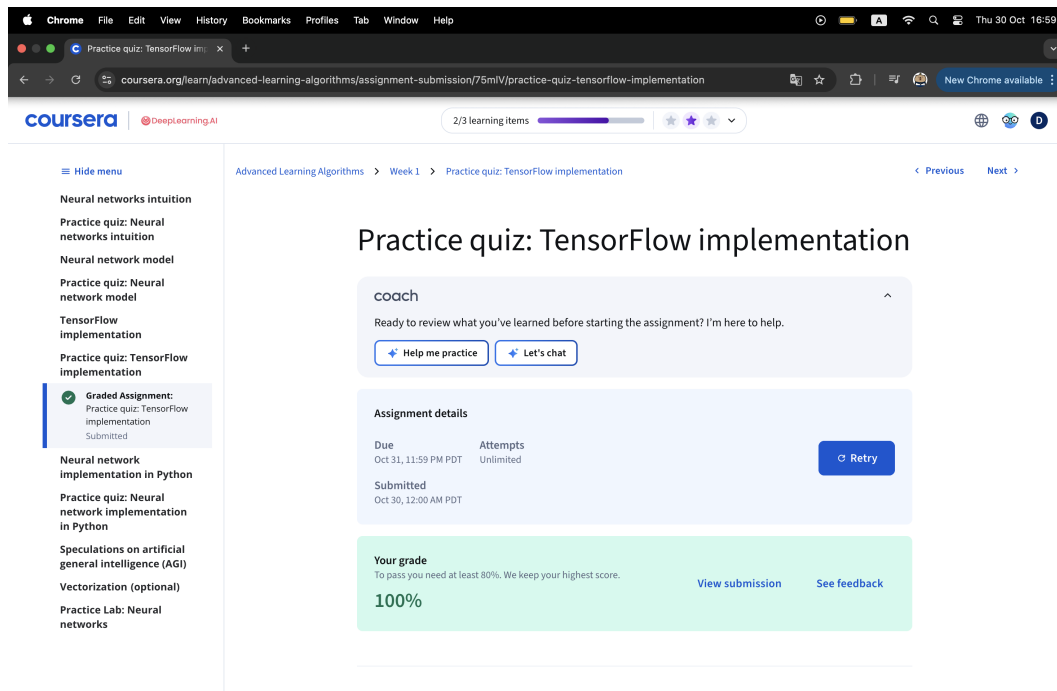


Figure 22: Completed Graded Assignment 3 (TensorFlow implementation)

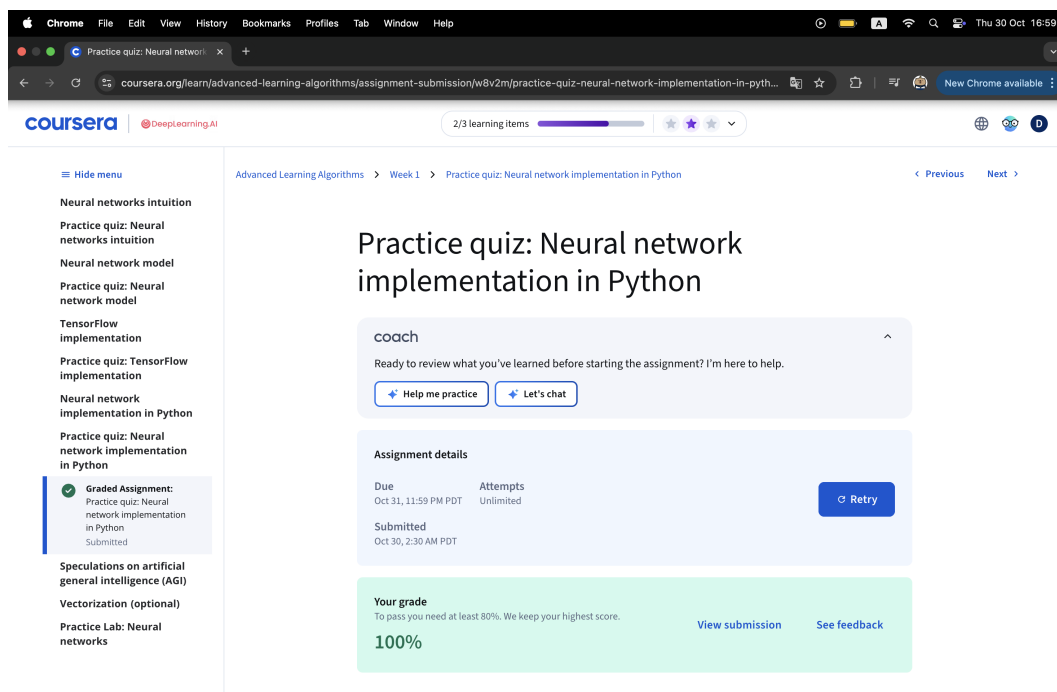


Figure 23: Completed Graded Assignment 4 (Neural network implementation in Python)

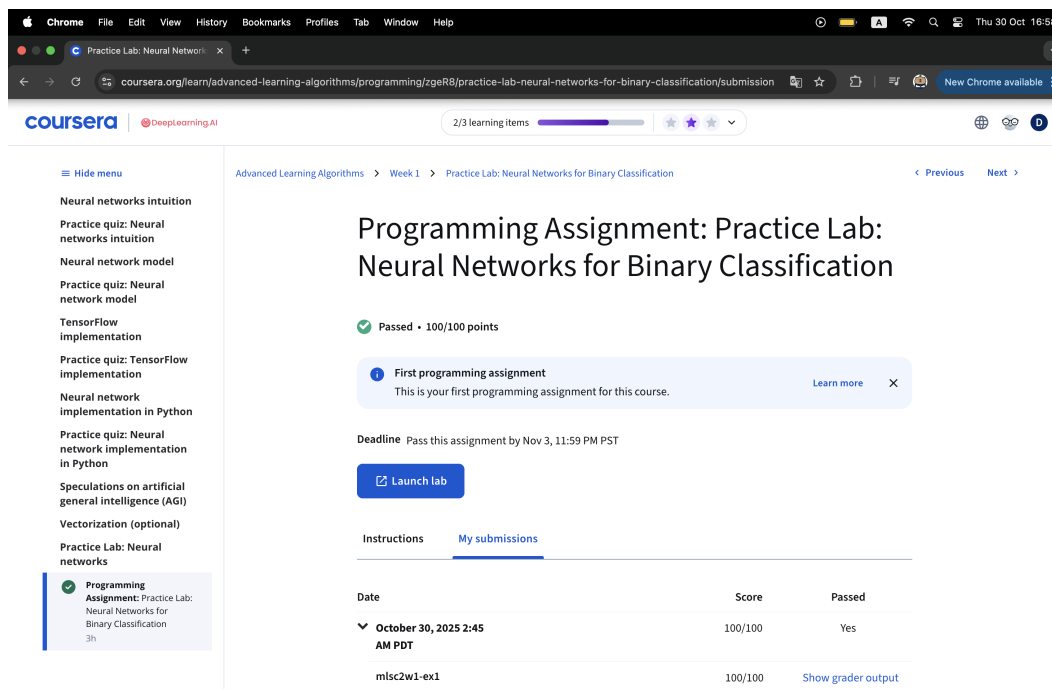


Figure 24: Completed Programming Assignment (Neural Networks for Binary Classification)

4.2 Week 2: Neural Network Training

4.2.1 Key Insights and Learnings

This module demystified the "black box" of training, focusing on the mathematical optimization and advanced techniques required for convergence.

- **Backpropagation Algorithm:** Mastered the chain rule application in computation graphs. This algorithm efficiently computes the partial derivatives ($\frac{\partial J}{\partial w}$) of the cost function, guiding the network on how to adjust weights to minimize error.
- **Advanced Optimization:**
 - Moved beyond standard Gradient Descent to **Adam (Adaptive Moment Estimation)**.
 - **Adam** adjusts learning rates individually for each parameter, resulting in faster convergence and better handling of local optima compared to standard SGD.
- **Handling Overfitting (Regularization):** Implemented strategies to improve model generalization:
 - **L2 Regularization:** Penalizing large weights in the cost function.
 - **Dropout:** Randomly deactivating neurons during training to prevent over-reliance on specific features.

4.2.2 Learning Evidence

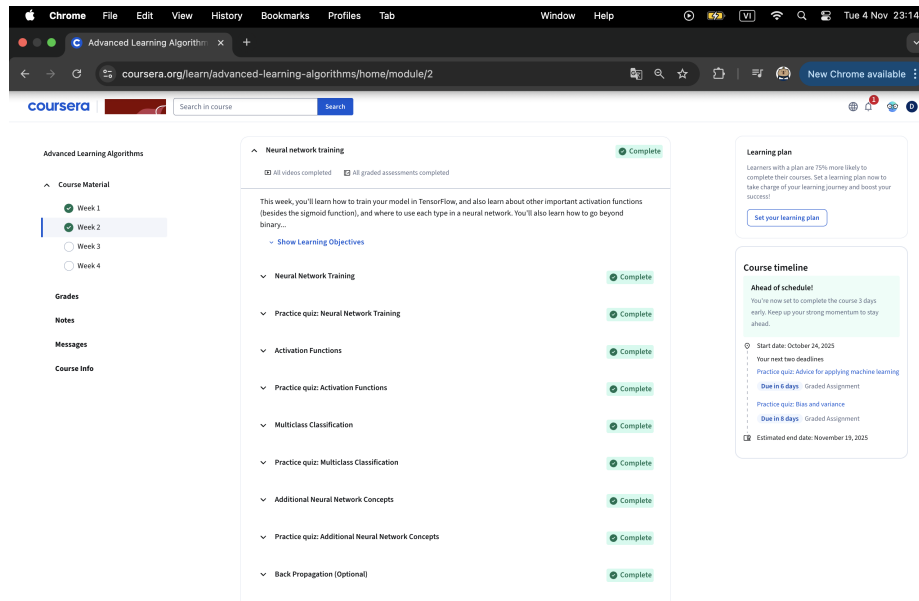


Figure 25: Lesson's completed date (04/11/2025)

4.2.3 Graded Assignment Evidence

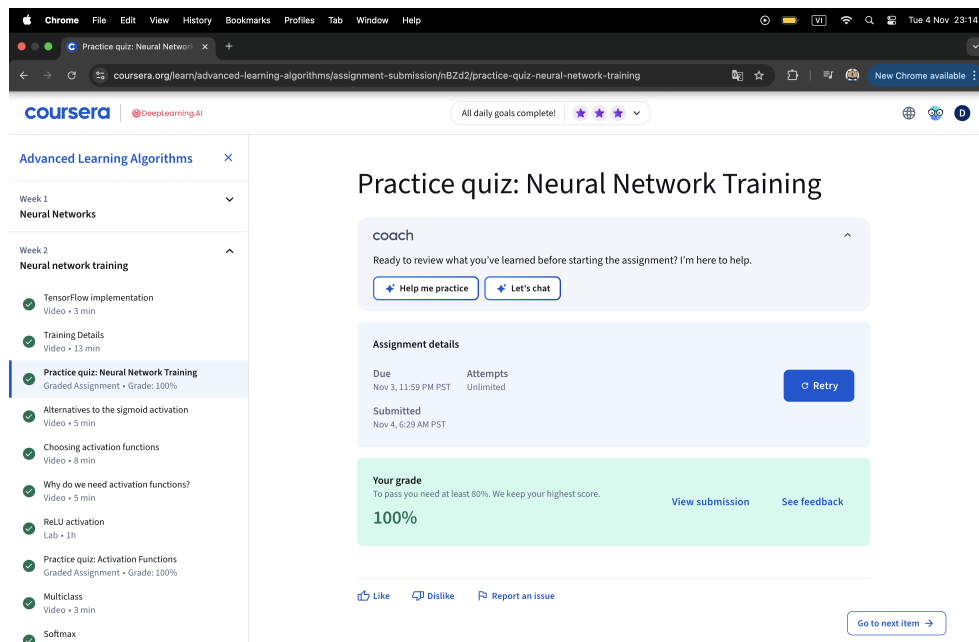


Figure 26: Completed Graded Assignment 1 (Neural Network Training)

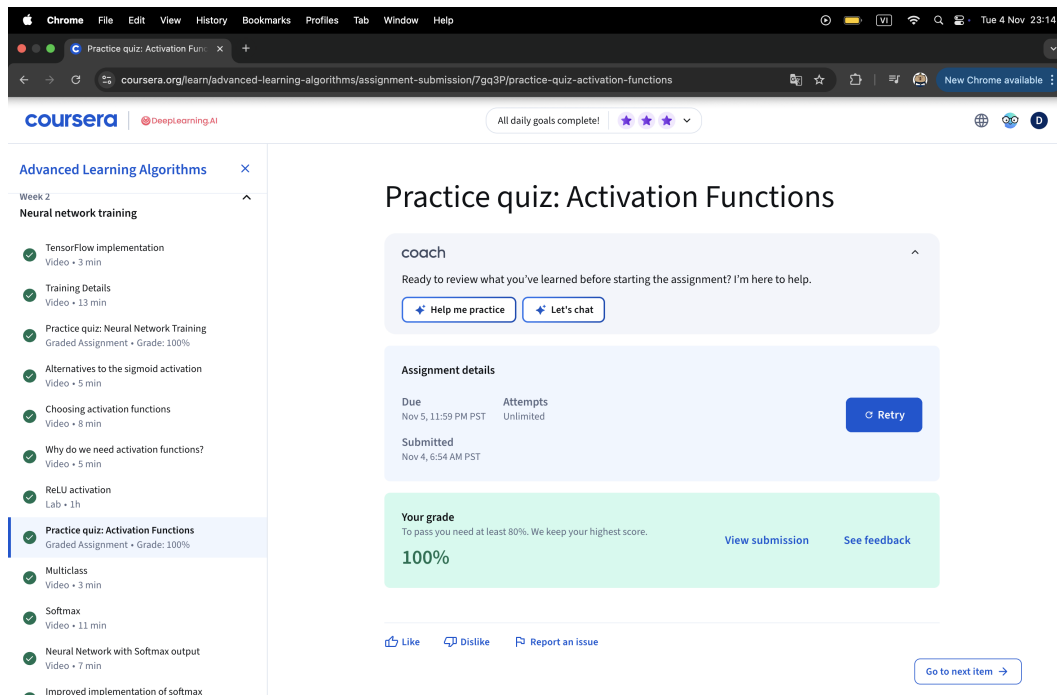


Figure 27: Completed Graded Assignment 2 (Activation Functions)

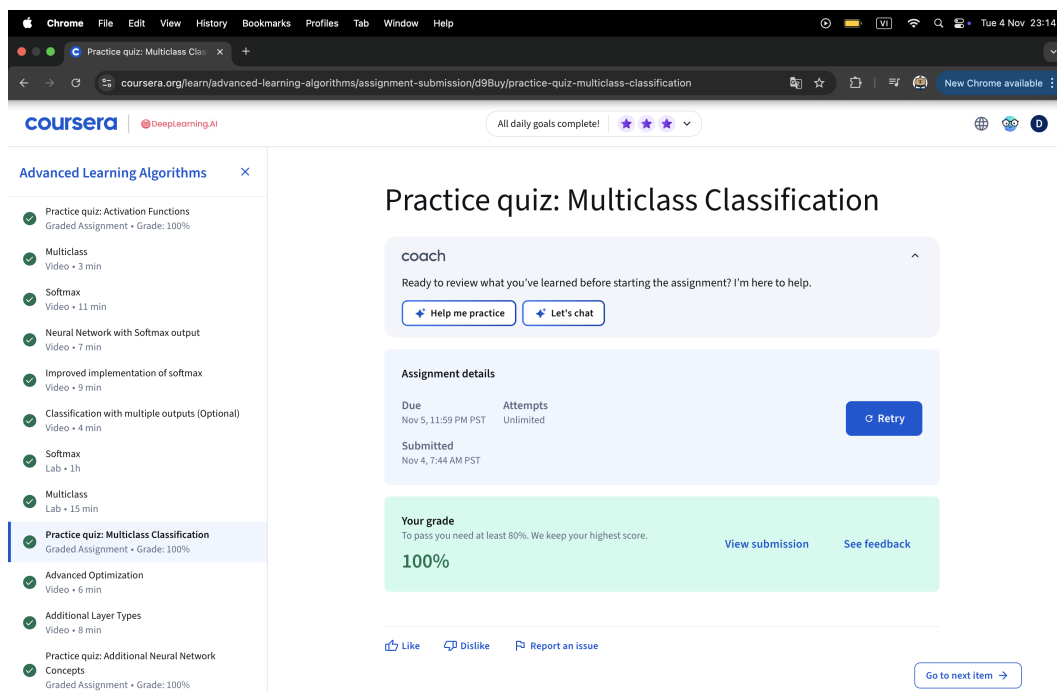


Figure 28: Completed Graded Assignment 3 (Multiclass Classification)

The screenshot shows the Coursera interface for the 'Advanced Learning Algorithms' course. The left sidebar lists various topics, with 'Practice quiz: Additional Neural Network Concepts' highlighted. The main content area displays the quiz title, a 'coach' message, and assignment details. The 'Your grade' section shows a score of 100%.

Practice quiz: Additional Neural Network Concepts

coach
Ready to review what you've learned before starting the assignment? I'm here to help.

[Help me practice](#) [Let's chat](#)

Assignment details

Due: Nov 7, 11:59 PM PST | Attempts: Unlimited | [Retry](#)

Submitted: Nov 4, 8:00 AM PST

Your grade
To pass you need at least 80%. We keep your highest score.

100% [View submission](#) [See feedback](#)

[Go to next item](#)

Figure 29: Completed Graded Assignment 4 (Additional Neural Network Concepts)

The screenshot shows the Coursera interface for the 'Advanced Learning Algorithms' course. The left sidebar lists various topics, with 'Practice Lab: Neural Networks for Multiclass classification' highlighted. The main content area displays the assignment title, a 'Passed' status, and a table of submissions.

Programming Assignment: Practice Lab: Neural Networks for Multiclass classification

✓ Passed • 100/100 points

Deadline Pass this assignment by Nov 7, 11:59 PM PST

[Launch lab](#)

Instructions **My submissions**

Date	Score	Passed
✓ November 4, 2025 8:12 AM PST	100/100	Yes
mLsc2w2_ex1	100/100	Show grader output

[Go to next item](#)

Figure 30: Completed Programming Assignment (Neural Networks for Multiclass classification)

4.3 Week 3: Advice for Applying Machine Learning

4.3.1 Key Insights and Learnings

Often considered the most valuable part of the specialization, this module provided a strategic framework for diagnosing and improving model performance.

- **Model Diagnostics (Bias vs. Variance):**
 - Utilized **Learning Curves** to visualize training and cross-validation errors.
 - **High Bias (Underfitting):** Addressed by increasing model complexity (adding layers/neurons) or reducing regularization.
 - **High Variance (Overfitting):** Addressed by collecting more data or adding regularization/dropout.
- **Error Metrics for Skewed Datasets:** Learned that accuracy is misleading for imbalanced classes (e.g., rare disease detection). Adopted **Precision**, **Recall**, and the **F1-Score** as robust evaluation metrics.
- **Iterative Loop of ML Development:** Embraced the cycle of Idea -> Code -> Experiment. The systematic split of data into **Train/Dev (Cross-Validation)/Test** sets is crucial for unbiased tuning.

4.3.2 Learning Evidence

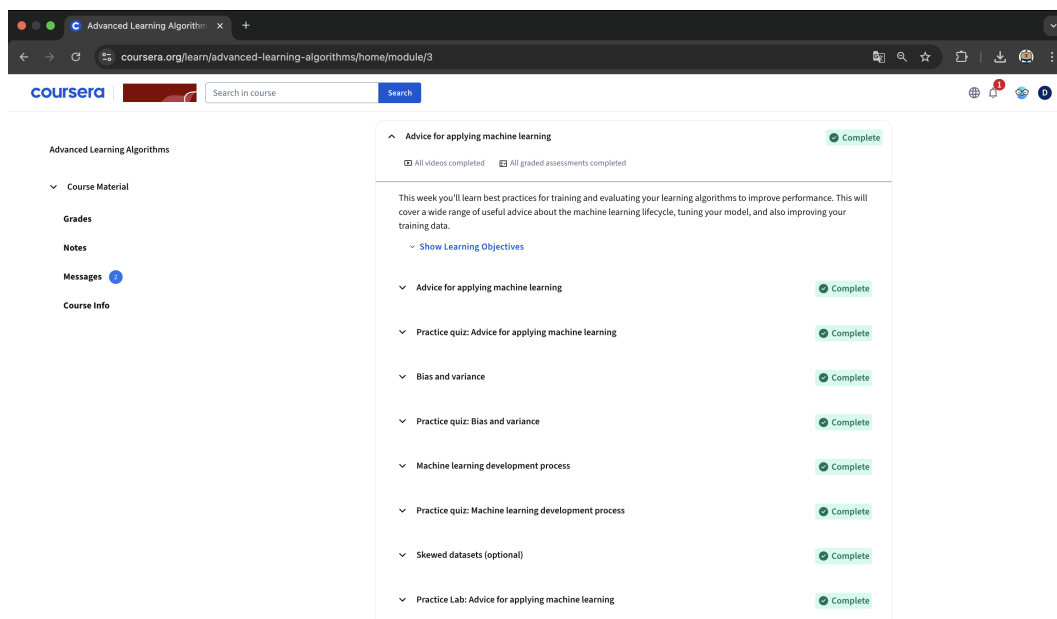


Figure 31: Lesson's completed date (07/11/2025)

4.3.3 Graded Assignment Evidence

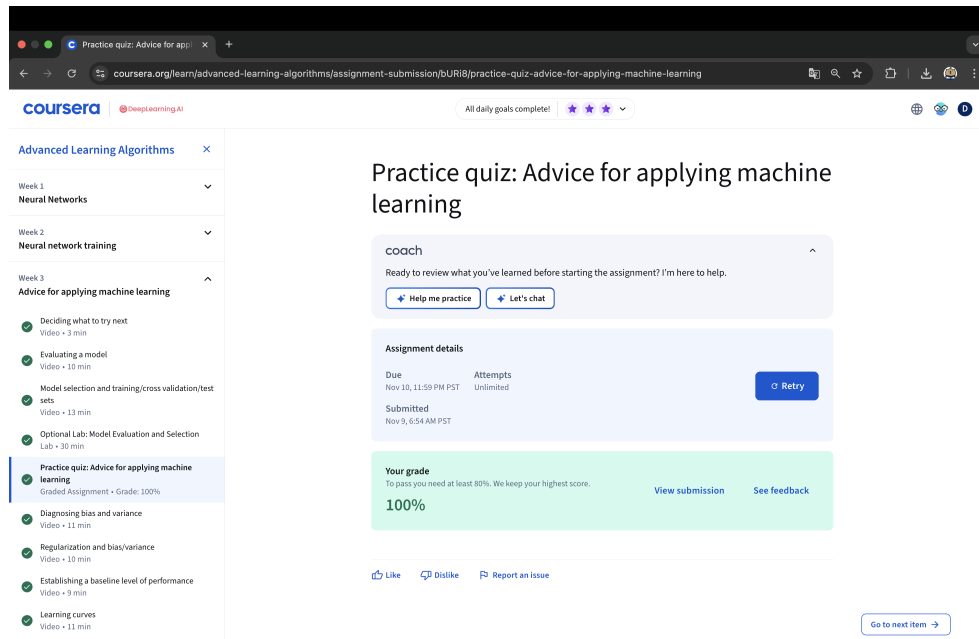


Figure 32: Completed Graded Assignment 1 (Advice for applying machine learning)

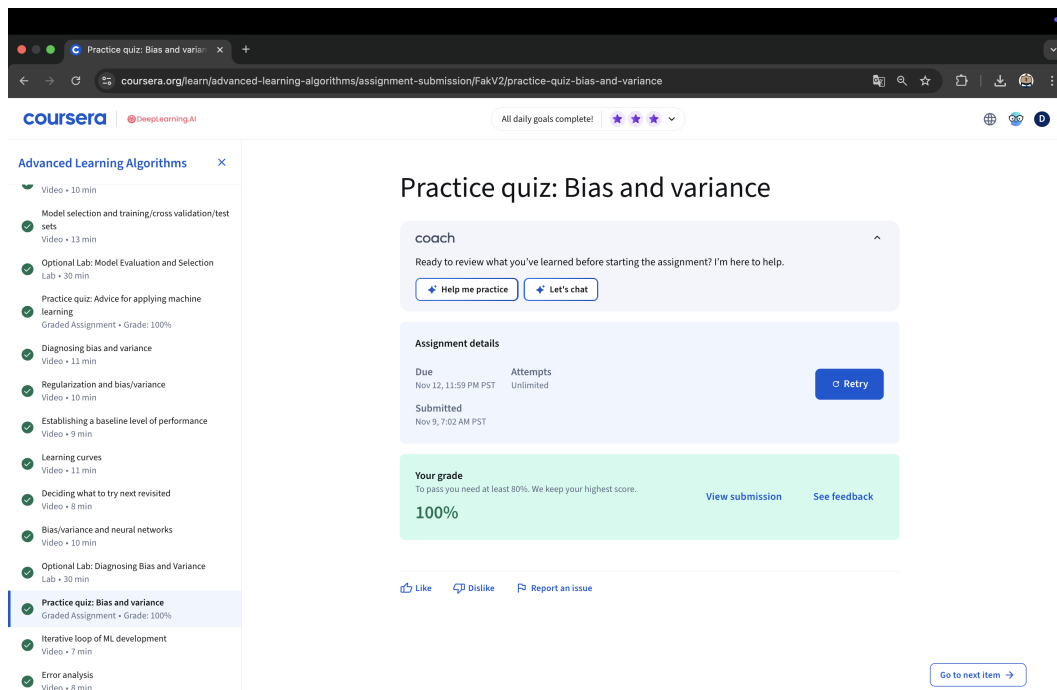


Figure 33: Completed Graded Assignment 2 (Bias and variance)

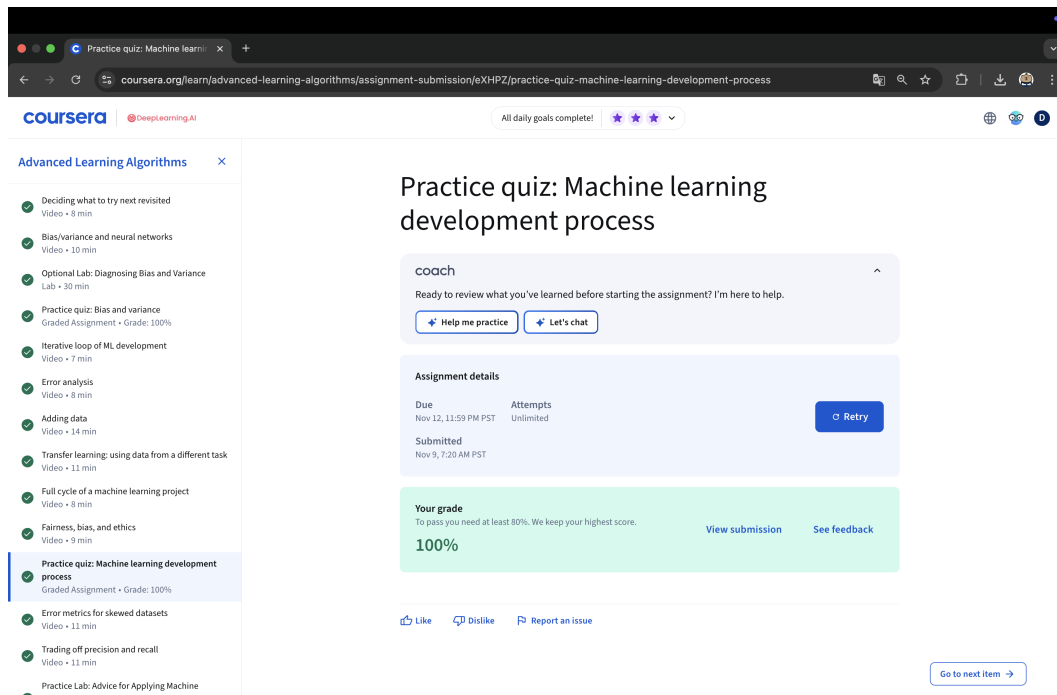


Figure 34: Completed Graded Assignment 3 (Machine learning development process)

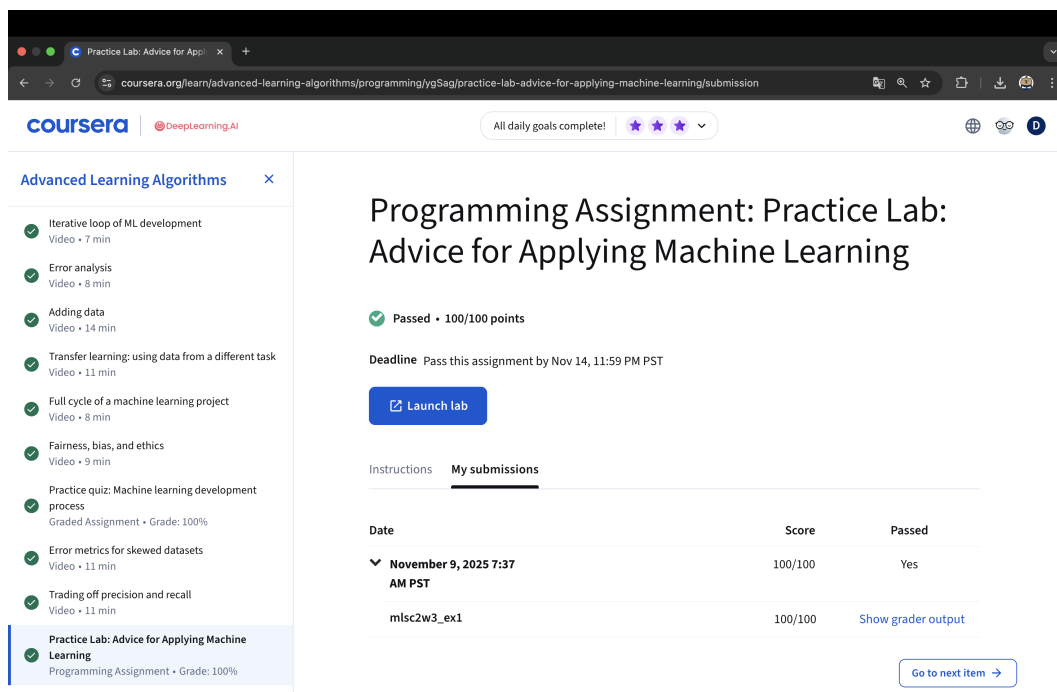


Figure 35: Completed Programming Assignment (Advice for Applying Machine Learning)

4.4 Week 4: Decision Trees and Ensemble Methods

4.4.1 Key Insights and Learnings

This module introduced robust alternatives to Neural Networks, particularly effective for structured (tabular) data.

- **Decision Trees Mechanics:**

- The learning process involves recursively splitting data to maximize **Information Gain** (reducing Entropy).
- Highly interpretable models where decision paths can be visualized and explained to non-technical stakeholders.

- **Ensemble Methods (The Wisdom of Crowds):**

- **Random Forests (Bagging):** Builds multiple trees on random subsets of data (sampling with replacement). It averages predictions to reduce **variance** and prevent overfitting.
- **XGBoost (Boosting):** A state-of-the-art algorithm that trains trees sequentially. Each new tree focuses on correcting the residual errors of the previous ones. It is currently the industry standard for tabular data competitions (e.g., Kaggle).

4.4.2 Learning Evidence

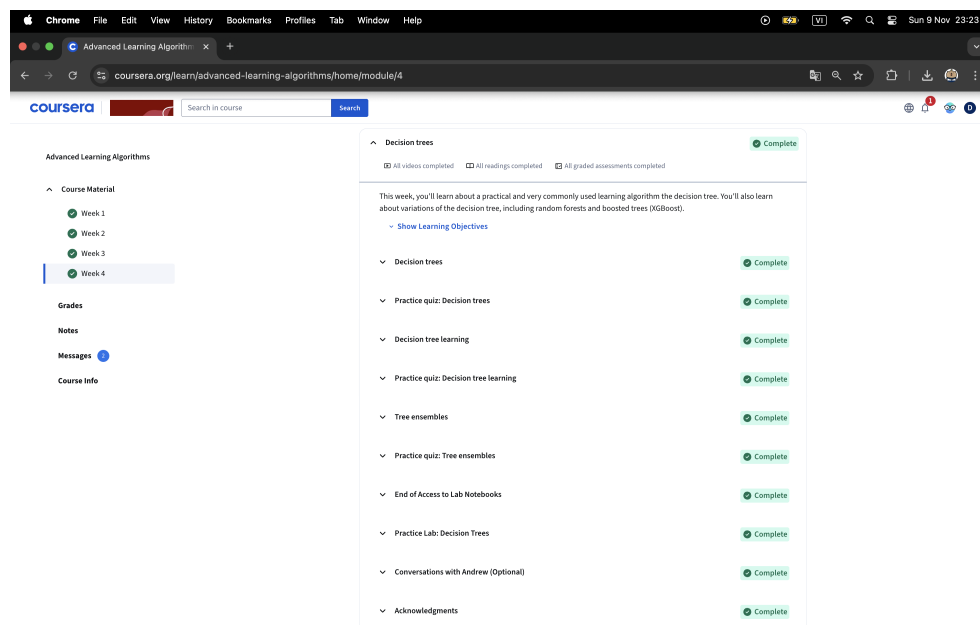


Figure 36: Lesson's completed date (09/11/2025)

4.4.3 Graded Assignment Evidence

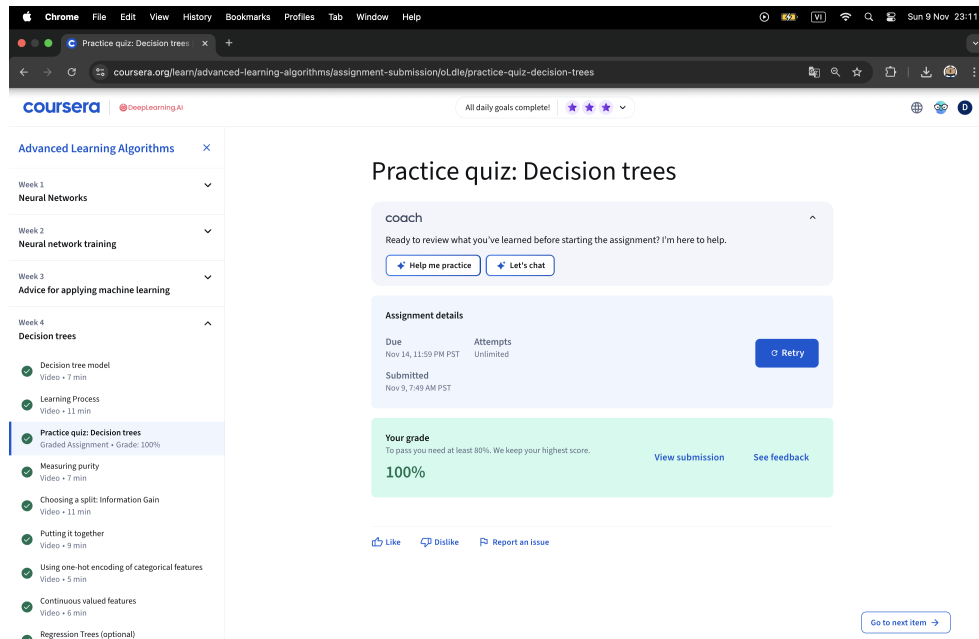


Figure 37: Completed Graded Assignment 1 (Decision trees)

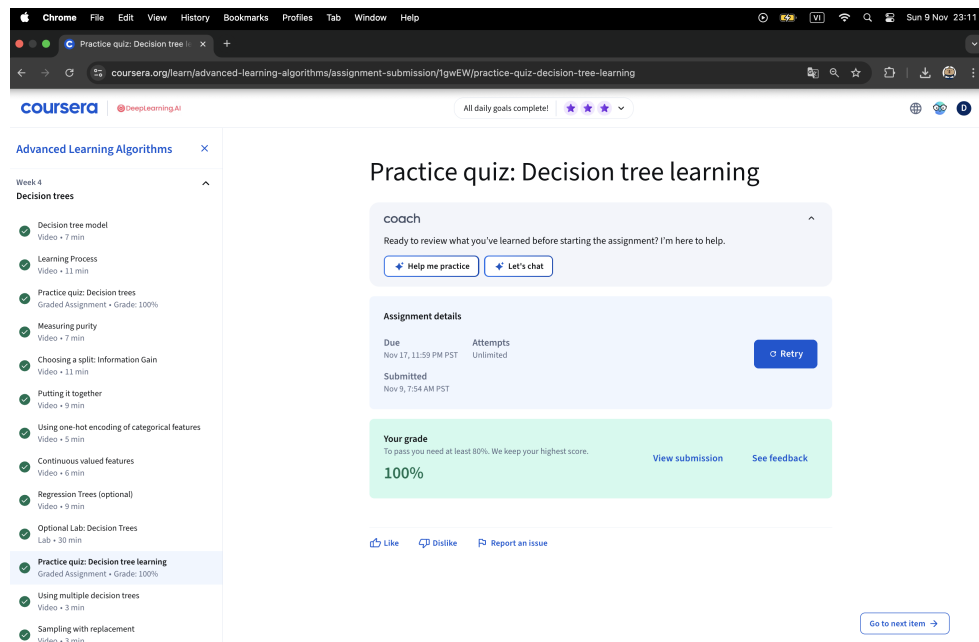


Figure 38: Completed Graded Assignment 2 (Decision tree learning)

The screenshot shows the Coursera interface for the 'Practice quiz: Tree ensembles' assignment. The left sidebar lists various topics under 'Advanced Learning Algorithms', with the current quiz highlighted. The main content area displays the quiz title, a 'coach' message, and assignment details. The 'Your grade' section shows a score of 100%.

Practice quiz: Tree ensembles

coach
Ready to review what you've learned before starting the assignment? I'm here to help.
[Help me practice](#) [Let's chat](#)

Assignment details

Due: Nov 17, 11:59 PM PST
Attempts: Unlimited
[Retry](#)

Submitted: Nov 9, 7:58 AM PST

Your grade
To pass you need at least 80%. We keep your highest score.
100%
[View submission](#) [See feedback](#)

[Like](#) [Dislike](#) [Report an issue](#)

[Go to next item](#)

Figure 39: Completed Graded Assignment 3 (Tree ensembles)

The screenshot shows the Coursera interface for the 'Programming Assignment: Practice Lab: Decision Trees'. The left sidebar lists various topics under 'Advanced Learning Algorithms', with the current lab highlighted. The main content area displays the lab title, a 'Passed' status, a deadline, and a table of submissions.

Programming Assignment: Practice Lab: Decision Trees

Passed • 100/100 points

Deadline Pass this assignment by Nov 19, 11:59 PM PST
[Launch lab](#)

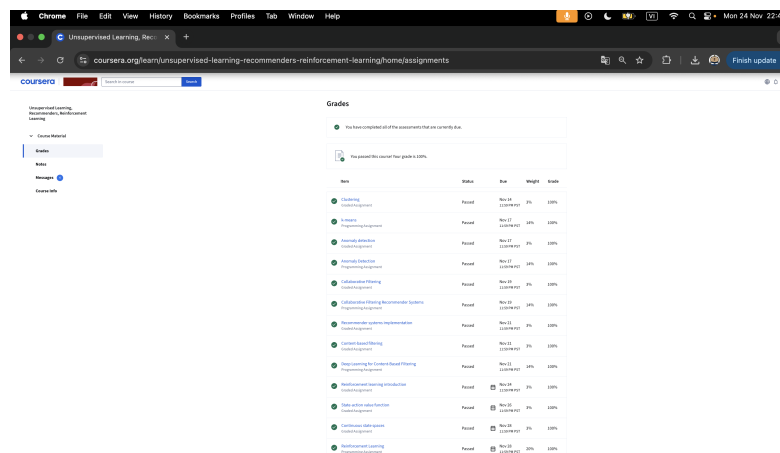
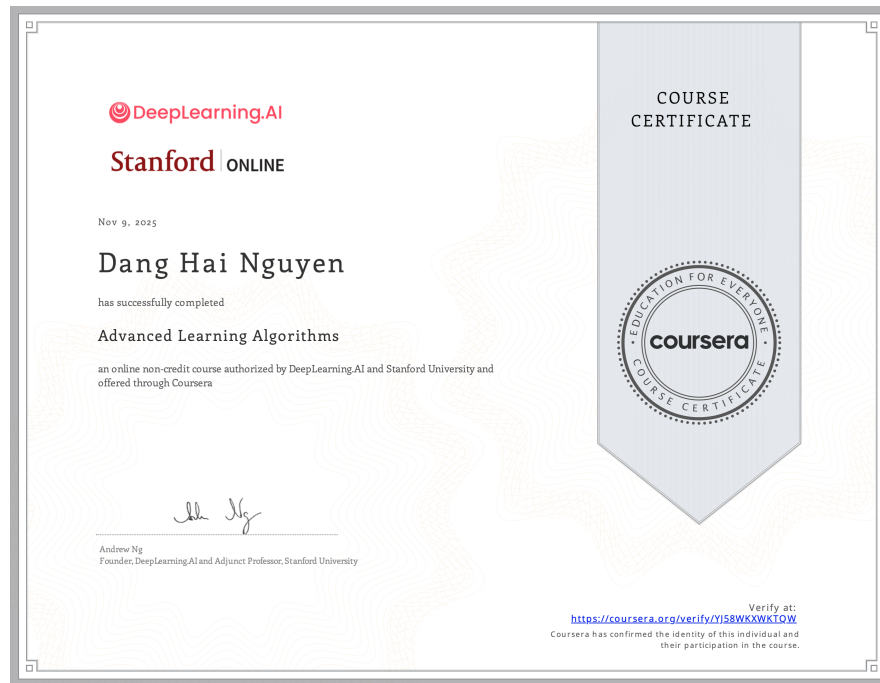
Instructions **My submissions**

Date	Score	Passed
November 9, 2025 8:02 AM PST	100/100	Yes
c2w4_ex1	100/100	Show grader output

[Like](#) [Dislike](#) [Report an issue](#)

[Go to next item](#)

Figure 40: Completed Programming Assignment (Decision Trees)



5.1 Week 1: Unsupervised Learning & Anomaly Detection

5.1.1 Key Insights and Learnings

This module focused on algorithms that derive structure from data without pre-existing labels, essential for exploratory data analysis, pattern recognition, and system monitoring.

- **Clustering with K-Means:** Learned to partition data into k distinct groups by minimizing the variance within each cluster.
 - Applied the *Elbow Method* to determine the optimal number of clusters (k) based on the distortion cost function.
 - **Applications:** Market segmentation, document organization, and astronomical data analysis.
- **Anomaly Detection:** Constructed probabilistic models using Gaussian (Normal) distribution to identify outliers or unusual events in a dataset.
 - **Technique:** Estimating parameters (μ, σ^2) to flag data points with very low probability density $(p(x) < \epsilon)$.
 - **Applications:** Fraud detection in finance, quality control in manufacturing, and server monitoring.
- **Dimensionality Reduction (PCA):** Utilized **Principal Component Analysis** to compress data while retaining critical information. By finding underlying axes of maximum variance, PCA reduces feature dimensionality, which aids in data visualization (2D/3D) and improves the efficiency of supervised learning models.

5.1.2 Learning Evidence

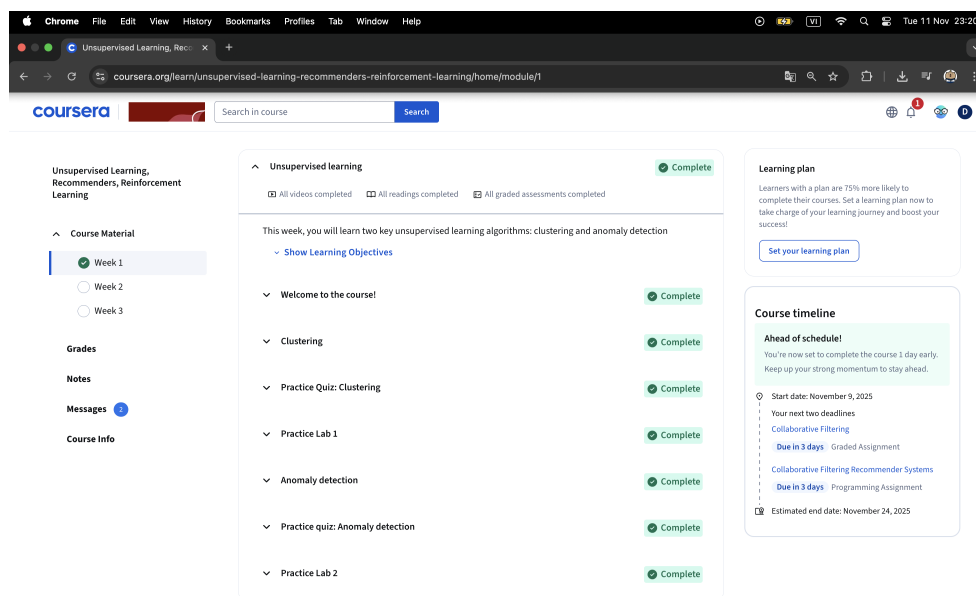
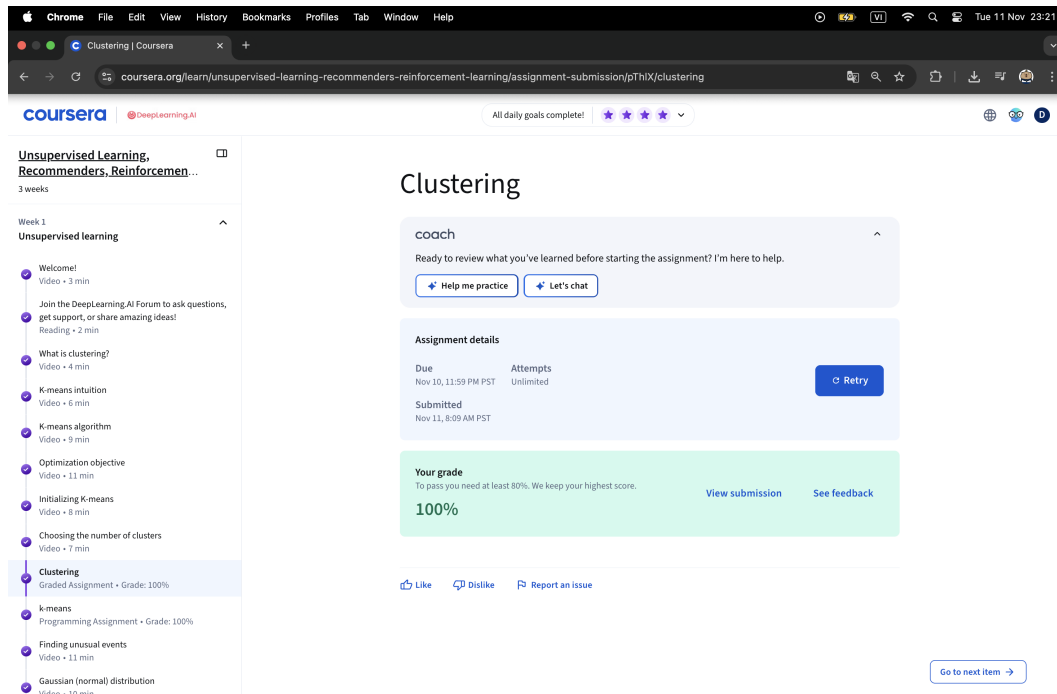


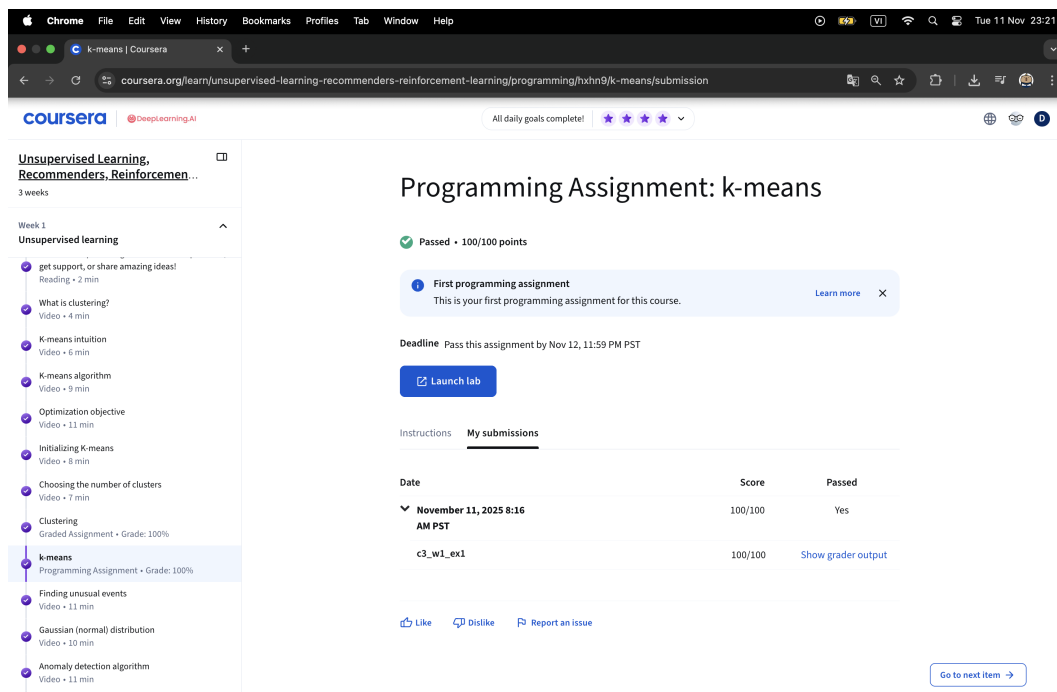
Figure 43: Lesson's completed date (11/11/2025)

5.1.3 Graded Assignment Evidence



The screenshot shows the Coursera interface for the 'Clustering' graded assignment. The left sidebar lists the course content, with 'Clustering' highlighted as the current assignment. The main content area displays the assignment title 'Clustering' and a 'coach' section with a message: 'Ready to review what you've learned before starting the assignment? I'm here to help.' Below this are buttons for 'Help me practice' and 'Let's chat'. The 'Assignment details' section shows the due date as 'Nov 10, 11:59 PM PST' and the submission date as 'Nov 11, 8:09 AM PST'. The 'Your grade' section shows a score of '100%' with a message: 'To pass you need at least 80%. We keep your highest score.' There are buttons for 'View submission' and 'See feedback'. At the bottom right, there is a 'Go to next item' button.

Figure 44: Completed Graded Assignment 1 (Clustering)



The screenshot shows the Coursera interface for the 'Programming Assignment: k-means'. The left sidebar lists the course content, with 'k-means' highlighted as the current assignment. The main content area displays the assignment title 'Programming Assignment: k-means' and a 'Passed' status with '100/100 points'. Below this is a 'First programming assignment' section with a message: 'This is your first programming assignment for this course.' The 'Deadline' section shows the due date as 'Nov 12, 11:59 PM PST' and a 'Launch lab' button. The 'My submissions' table shows the following data:

Date	Score	Passed
November 11, 2025 8:16 AM PST	100/100	Yes
c3_w1_ex1	100/100	Show grader output

At the bottom right, there is a 'Go to next item' button.

Figure 45: Completed Programming Assignment (k-means)

The screenshot shows the Coursera interface for the 'Anomaly detection' assignment. The left sidebar lists the course content, including 'Week 1: Unsupervised learning' and 'Week 2: Recommender systems'. The main content area displays the assignment title 'Anomaly detection' and a 'coach' message. Below this, the 'Assignment details' section shows the due date as 'Nov 12, 11:59 PM PST' and the submission date as 'Nov 11, 8:12 AM PST'. The 'Your grade' section indicates a score of '100%' with a message: 'To pass you need at least 80%. We keep your highest score.' There are buttons for 'View submission' and 'See feedback'. At the bottom, there are buttons for 'Like', 'Dislike', and 'Report an issue', and a 'Go to next item' button.

Figure 46: Completed Graded Assignment 3 (Anomaly detection)

The screenshot shows the Coursera interface for the 'Programming Assignment: Anomaly Detection'. The left sidebar lists the course content, including 'Week 1: Unsupervised learning' and 'Week 2: Recommender systems'. The main content area displays the assignment title 'Programming Assignment: Anomaly Detection' and a 'Passed • 100/100 points' status. Below this, the 'Deadline' section shows the due date as 'Nov 12, 11:59 PM PST' and a 'Launch lab' button. The 'My submissions' table shows the submission details:

Date	Score	Passed
November 11, 2025 8:19 AM PST	100/100	Yes
c3_w1_ex2	100/100	Show grader output

At the bottom, there are buttons for 'Like', 'Dislike', and 'Report an issue', and a 'Go to next item' button.

Figure 47: Completed Programming Assignment (Anomaly Detection)

5.2 Week 2: Recommender Systems

5.2.1 Key Insights and Learnings

This module demystified the architecture behind modern personalization engines used by platforms like Netflix and Amazon, implementing both content-based and interaction-based approaches.

- **Collaborative Filtering:** Implemented algorithms that predict user preferences based on community behavior. The model learns that users who agreed in the past will likely agree in the future, without needing explicit item attributes.
- **Content-Based Filtering:** Developed models that recommend items based on the similarity between item features (e.g., genre, keywords) and a user's profile.
- **Matrix Factorization & Embeddings:**
 - Learned to represent users and items as vectors in a lower-dimensional space.
 - The dot product of these vectors predicts the user rating ($y^{(i,j)} = w^{(j)} \cdot x^{(i)} + b$).
 - Utilized **Mean Normalization** to handle the "Cold Start" problem for new users with no history.

5.2.2 Learning Evidence

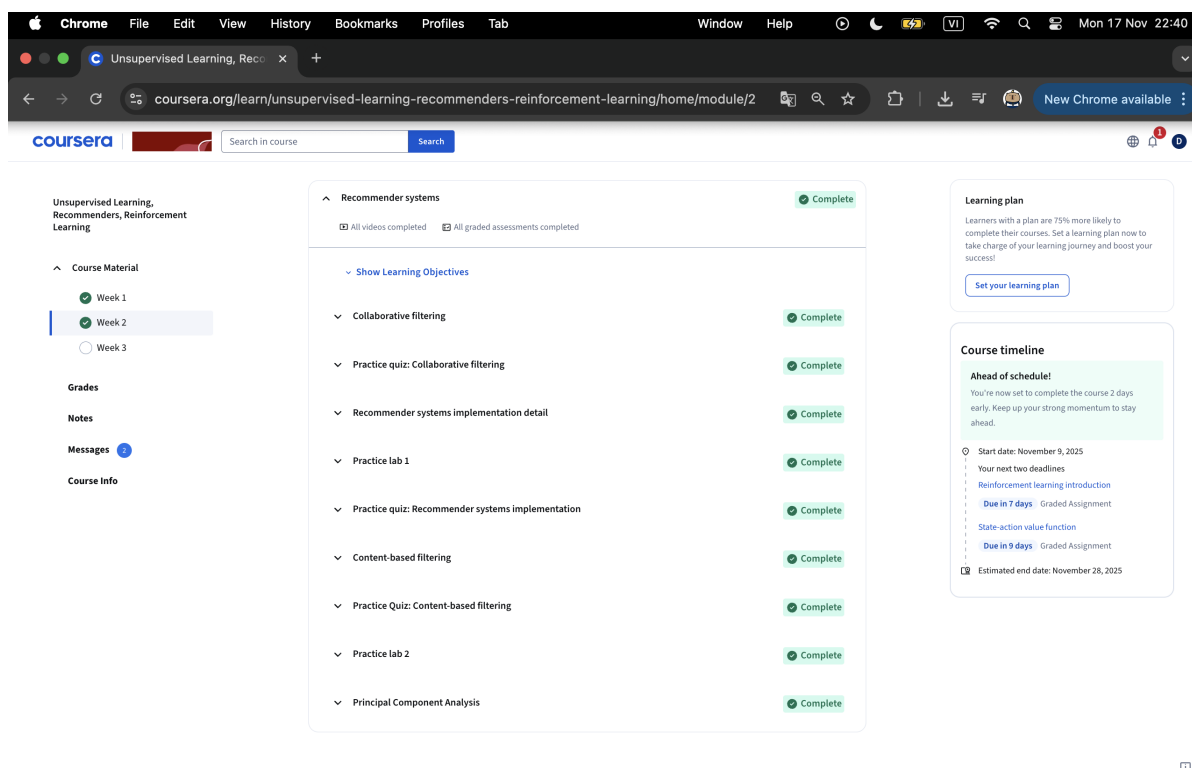


Figure 48: Lesson's completed date (17/11/2025)

5.2.3 Graded Assignment Evidence

The screenshot shows a web browser window displaying a Coursera assignment page. The browser's address bar shows the URL: `coursera.org/learn/unsupervised-learning-recommenders-reinforcement-learning/assignment-submis...`. The page title is "Collaborative Filtering". On the left sidebar, under "Unsupervised Learning, Recommenders, Reinforcement Learning", the "Collaborative Filtering" assignment is highlighted with a grade of 100%. The main content area shows "Assignment details" with a due date of Nov 19, 11:59 PM PST, unlimited attempts, and a submission date of Nov 17, 7:01 AM PST. A "Retry" button is visible. Below this, a green box displays "Your grade" as 100%, with a note: "To pass you need at least 80%. We keep your highest score." Links for "View submission" and "See feedback" are provided. At the bottom, there are "Like", "Dislike", and "Report an issue" buttons, and a "Go to next item" button.

Figure 49: Completed Graded Assignment 1 (Collaborative Filtering)

The screenshot shows a web browser window displaying a Coursera assignment page. The browser's address bar shows the URL: `coursera.org/learn/unsupervised-learning-recommenders-reinforcement-learning/programming/TL...`. The page title is "Programming Assignment: Collaborative Filtering Recommender Systems". On the left sidebar, under "Unsupervised Learning, Recommenders, Reinforcement Learning", the "Collaborative Filtering Recommender Systems" programming assignment is highlighted with a grade of 100%. The main content area shows a green checkmark indicating "Passed • 100/100 points". The deadline is "Pass this assignment by Nov 19, 11:59 PM PST". A "Launch lab" button is visible. Below this, there are tabs for "Instructions" and "My submissions". The "My submissions" tab is active, showing a table of submissions:

Date	Score	Passed
November 17, 2025 7:29 AM PST	100/100	Yes
c3_w2_ex1	100/100	Show grader output

At the bottom, there is a "Go to next item" button.

Figure 50: Completed Programming Assignment (Collaborative Filtering Recommender Systems)

The screenshot shows a web browser window with the Coursera website. The page title is "Recommender systems implementation". On the left, a sidebar lists the course content for "Week 2: Recommender systems", with "Recommender systems implementation" highlighted. The main content area shows "Assignment details" with a due date of Nov 21, 11:59 PM PST and unlimited attempts. Below this, the "Your grade" section displays "100%" with a note: "To pass you need at least 80%. We keep your highest score." There are links for "View submission" and "See feedback". At the bottom, there are buttons for "Like", "Dislike", "Report an issue", and "Go to next item".

Figure 51: Completed Graded Assignment 3 (Recommender systems implementation)

The screenshot shows a web browser window with the Coursera website. The page title is "Content-based filtering". On the left, a sidebar lists the course content for "Week 2: Recommender systems", with "Content-based filtering" highlighted. The main content area shows "Assignment details" with a due date of Nov 21, 11:59 PM PST and unlimited attempts. Below this, the "Your grade" section displays "100%" with a note: "To pass you need at least 80%. We keep your highest score." There are links for "View submission" and "See feedback". At the bottom, there are buttons for "Like", "Dislike", "Report an issue", and "Go to next item".

Figure 52: Completed Graded Assignment 3 (Content-based filtering)

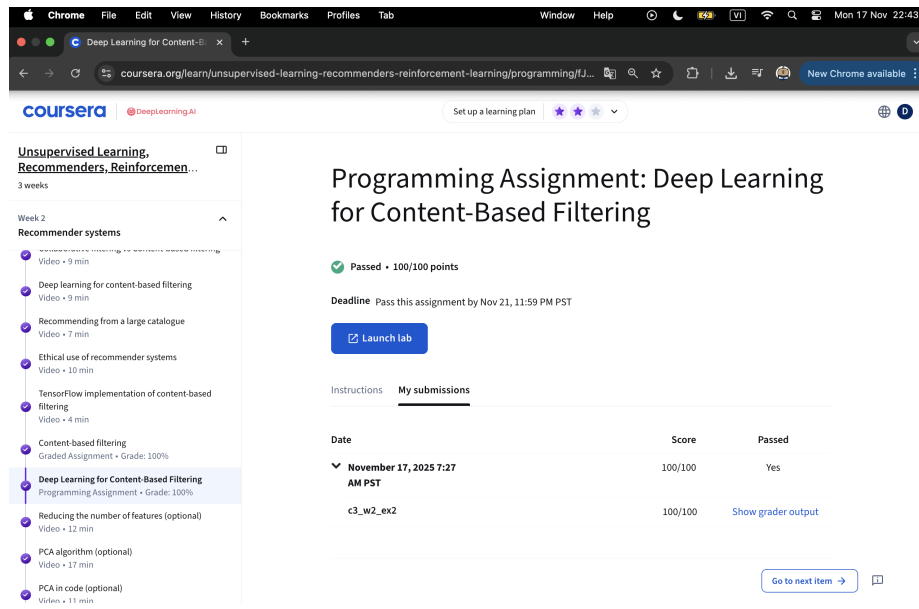


Figure 53: Completed Programming Assignment (Deep Learning for Content-Based Filtering)

5.3 Week 3: Reinforcement Learning (RL)

5.3.1 Key Insights and Learnings

This module introduced a paradigm shift where an agent learns a policy to maximize rewards over time, rather than minimizing instantaneous error.

- **The RL Framework (MDP):** Defined the problem as a Markov Decision Process, consisting of States (S), Actions (A), Rewards (R), and a Discount factor (γ). The goal is to maximize the expected *Return*.
- **The Bellman Equation:** Understood the recursive relationship that defines the value of a state-action pair. This equation is the foundation for updating the Q-function to estimate long-term rewards.
- **Deep Q-Learning (DQN):**
 - Replaced traditional Q-tables with Neural Networks to approximate the Q-function, enabling the agent to handle complex, high-dimensional state spaces (e.g., video game pixels or robot sensors).
 - **Exploration vs. Exploitation:** Implemented the ϵ -greedy policy to balance trying new actions versus leveraging known optimal actions.

5.3.2 Learning Evidence

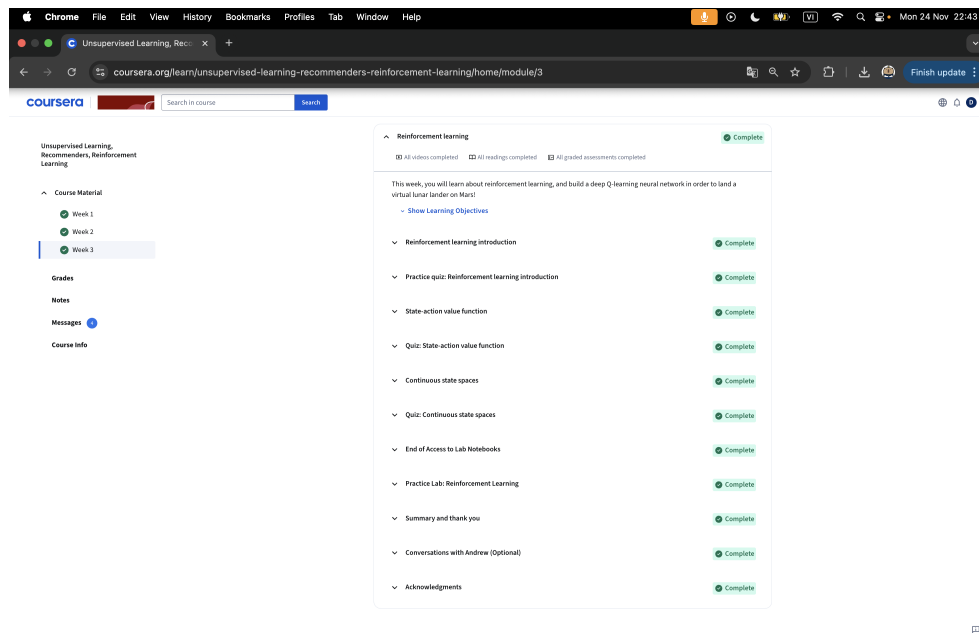


Figure 54: Lesson's completed date (24/11/2025)

5.3.3 Graded Assignment Evidence

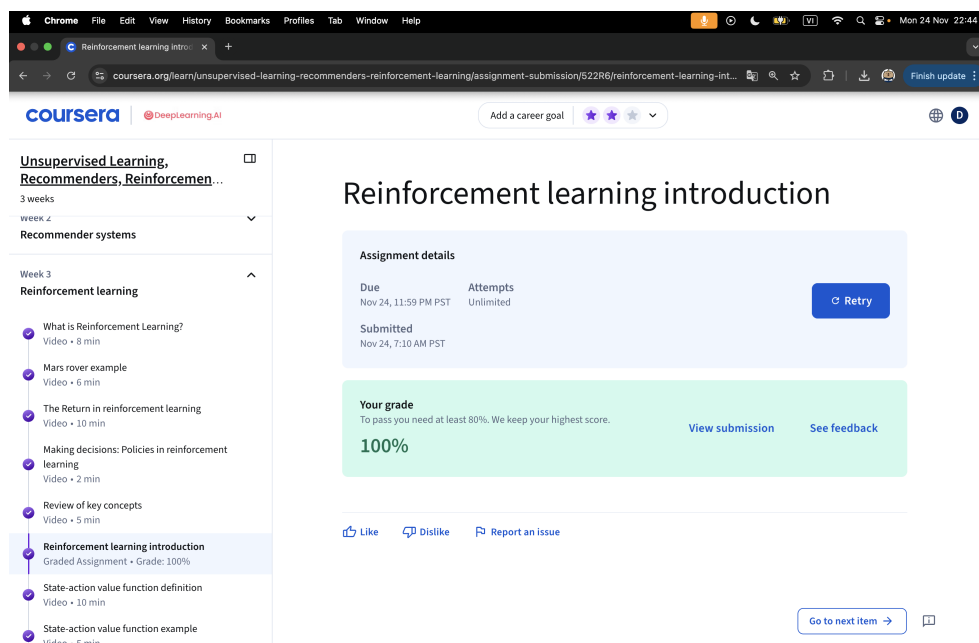


Figure 55: Completed Graded Assignment 1 (Reinforcement learning introduction)

The screenshot shows the Coursera interface for the 'State-action value function' assignment. The left sidebar lists the course content, with 'State-action value function' highlighted under 'Reinforcement learning'. The main content area displays the assignment details, including the due date (Nov 26, 11:59 PM PST), attempts (Unlimited), and submitted date (Nov 24, 7:13 AM PST). A 'Retry' button is visible. Below this, the 'Your grade' section shows a score of 100%, with a note that the user needs at least 80% to pass. Links for 'View submission' and 'See feedback' are provided. At the bottom, there are buttons for 'Like', 'Dislike', 'Report an issue', and 'Go to next item'.

State-action value function

Assignment details

Due Nov 26, 11:59 PM PST Attempts Unlimited [Retry](#)

Submitted Nov 24, 7:13 AM PST

Your grade To pass you need at least 80%. We keep your highest score. [View submission](#) [See feedback](#)

100%

[Like](#) [Dislike](#) [Report an issue](#)

[Go to next item](#)

Figure 56: Completed Graded Assignment 2 (State-action value function)

The screenshot shows the Coursera interface for the 'Continuous state spaces' assignment. The left sidebar lists the course content, with 'Continuous state spaces' highlighted under 'Reinforcement learning'. The main content area displays the assignment details, including the due date (Nov 26, 11:59 PM PST), attempts (Unlimited), and submitted date (Nov 24, 7:15 AM PST). A 'Retry' button is visible. Below this, the 'Your grade' section shows a score of 100%, with a note that the user needs at least 80% to pass. Links for 'View submission' and 'See feedback' are provided. At the bottom, there are buttons for 'Go to next item' and a bookmark icon.

Continuous state spaces

Assignment details

Due Nov 26, 11:59 PM PST Attempts Unlimited [Retry](#)

Submitted Nov 24, 7:15 AM PST

Your grade To pass you need at least 80%. We keep your highest score. [View submission](#) [See feedback](#)

100%

[Go to next item](#)

Figure 57: Completed Graded Assignment 3 (Continuous state spaces)

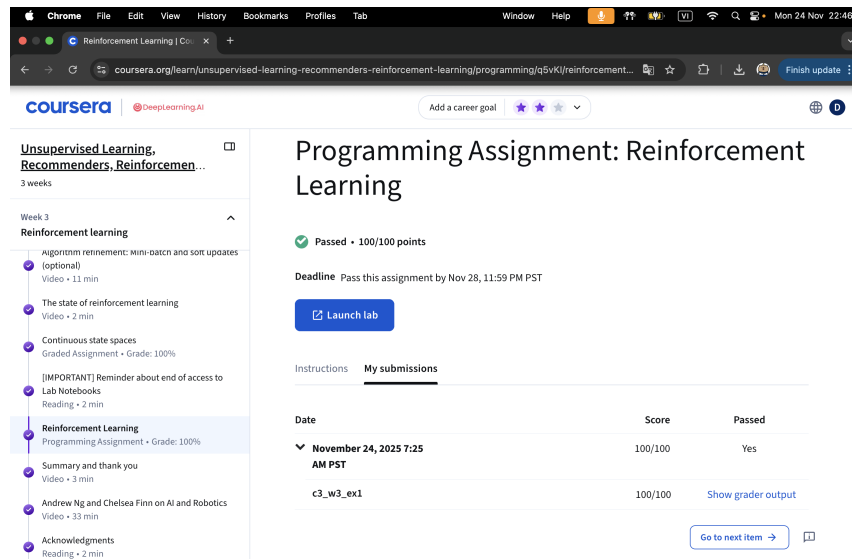


Figure 58: Completed Programming Assignment (Reinforcement Learning)

5.4 Certificate for Completion

Figure 59: Certificate of completion for the course *Unsupervised Learning, Recommenders, and Reinforcement Learning*.

6 Conclusion: A Comprehensive Journey into Machine Learning

The completion of the *Coursera Machine Learning Specialization* has been a transformative educational experience. This three-course program provided a comprehensive and brilliantly structured journey, starting from the foundational mathematics of supervised learning and culminating in an exploration of advanced, state-of-the-art concepts.

The specialization began by building a solid foundation in **Supervised Learning**, demystifying the core mechanics of regression and classification. It then progressed to **Advanced Learning Algorithms**, where I gained the practical skills to build and train powerful neural networks and learned indispensable, real-world methodologies for diagnosing and improving model performance. The final course, **Unsupervised Learning, Recommenders, and Reinforcement Learning**, expanded my perspective beyond labeled data, introducing me to the exciting possibilities of discovering hidden patterns, building personalized systems, and training intelligent agents.

Throughout this entire curriculum, the emphasis was not just on theoretical knowledge but also on practical application. Through hands-on labs using Python, NumPy, and TensorFlow, I translated complex mathematical concepts into functional code. More importantly, I acquired a robust mental framework for approaching machine learning problems systematically—from data preparation and feature engineering to model selection, training, and evaluation.

This specialization has equipped me with a skill set that is both broad and deep, giving me the confidence to tackle real-world data challenges and pursue more advanced topics in artificial intelligence. It has been an invaluable step in my academic and professional development.

7 Certificate of Completion

As the primary evidence of my achievement, I have successfully completed all required coursework, programming assignments, and quizzes for the entire three-course *Machine Learning Specialization*.

The certificate is issued by Coursera in partnership with DeepLearning.AI and Stanford Online, verifying my proficiency in the topics covered throughout this report.

7.1 Specialization Certificate Image



Figure 60: Official Certificate of Completion for the Machine Learning Specialization.

7.2 Verifiable Certificate Link

The certificate can be officially verified by following the URL below:

Course Certificates Completed